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ACTION-CENTERED CAUSAL BANDITS FOR BUDGET-CONSTRAINED PROMOTION TARGETING

Yifan Lin **Kaixu Yang** **Di Mo**
LinkedIn Corporation
Mountain View, CA, USA
{yiflin,kayang,dmo}@linkedin.com

ABSTRACT

Web platforms select among costly interventions for each user to maximize incremental outcomes under a global budget. We formulate this problem as a budget-constrained contextual bandit with heterogeneous treatment effects. Although an action-independent outcome baseline cancels from treatment comparisons, modeling a nonlinear baseline can increase statistical complexity.

We propose DR-CATE-CBwK, an action-centered causal bandit that directly learns treatment contrasts. Its online update remains valid under baseline-model misspecification without inverse-propensity weighting. It combines an offline doubly robust learner to warm-start treatment-effect estimates, an online action-centered estimator for sequential contrast learning, and a primal-dual controller with an affordable-action rule to enforce the effective-cost budget pathwise.

For linear conditional average treatment effects (CATEs) in a stationary i.i.d. setting, we prove a high-probability ($\widetilde{O}(Kd\sqrt{T})$) regret bound, where K is the number of priced actions, d the context dimension, and T the horizon. We benchmark against the fluid linear program (LP) under the coverage-guaranteed action rule. We further show that the action-centered estimating moment remains unbiased under any predictable baseline estimate, whereas a specified separate-arm reward regression can exhibit asymptotic contrast bias under action-dependent logging and shared-baseline misspecification.

On a semi-synthetic job-marketplace benchmark with real logged features and simulated outcomes, our practical Thompson-restricted variant achieves the lowest regret among evaluated budget-feasible methods in the operating regime. It reduces regret by approximately 20% versus the strongest reward-based baseline, outperforming raw-reward contextual bandits, BwK baselines, and a static offline policy, while never exceeding the effective-cost budget.

Keywords contextual bandits · causal inference · semiparametric bandits · uplift modeling · bandits with knapsacks · budget-constrained optimization

1 Introduction

Web platforms repeatedly choose, for each user encounter, one of several costly interventions to maximize an incremental outcome under a global budget: promotional credits, retention discounts, coupons, or ad subsidies. Candidate actions differ in cost, their effects differ across users, and total spend is bounded over a horizon.

The standard tool is a contextual bandit [1, 2] that models the conditional reward $\mathbb{E}[Y \mid X = x, A = a] = f(x) + \tau_a(x)$: an action-independent baseline f plus the conditional average treatment effect (CATE) $\tau_a(x) = \mathbb{E}[Y(a) - Y(a_0) \mid X = x]$ against a no-intervention control a_0. This decomposition does not change the objective: for every policy π,

$$\mathbb{E}_{X,A\sim\pi}[Y(A)] = \mathbb{E}_X[f(X)] + \mathbb{E}_{X,A\sim\pi}[\tau_A(X)], \tag{1}$$

and the first term is policy-independent, so a correctly specified reward-based and a CATE-based bandit with knapsacks (CBwK) share the same constrained optimum. The difference is statistical: a learner that fits raw conditional rewards also models the possibly nonlinear nuisance f that cancels from every action comparison. This can increase estimation variance or, under a misspecified reward class, create spurious contrasts when separate action models see different context distributions.

A budget makes this distinction consequential because scarce spend must be ranked by incremental value. If only 20% of traffic can be funded, a \$100 credit with a $+3$ percentage-point uplift on high-baseline power posters but a $+15$ point uplift on persuadable mid-tier posters should go to the latter, a choice a naive response-propensity ranking that does not compare treatment against control would miss. The question is how to learn the treatment contrasts reliably without first fitting f.

We introduce DR-CATE-CBwK, an online algorithm that learns these contrasts under a hard budget. Building on action-centered bandits [3, 4], we couple a baseline-robust estimator with an offline doubly robust CATE warm start and a primal-dual controller with an affordable-action restriction. The warm start enters the confidence radius as an initial estimation error; a valid external radius can tighten its contribution to regret (Assumption 9). Our contribution is the joint guarantee for this offline-to-online budgeted learning procedure.

Our contributions are:

(1) DR-CATE-CBwK (Algorithm 1), combining offline causal initialization, online semiparametric contrast learning, and pathwise budget feasibility. Its farthest-pair action rule supplies active-set coverage without a permanent exploration floor.

(2) A high-probability $\widetilde{O}(Kd\sqrt{T})$ regret bound against the fluid LP for linear CATEs under stationary i.i.d. contexts and the stated noise, coverage, and confidence conditions (Theorem 1). We also characterize the baseline robustness of the centered moment and the bias of a specified separate-arm reward regression (Theorem 2).

(3) A semi-synthetic job-promotion evaluation with real logged features and simulated outcomes. The practical Thompson-restricted variant has regret 498 versus 620 for Raw-Flex-CBwK, the strongest non-ablation baseline, approximately 20% lower in the operating regime. Comparisons with the farthest-pair and coverage-mixture variants examine the effect of the action-selection rule on regret. Additional studies evaluate baseline misspecification, offline-to-online shift, nonlinear CATEs, and budget feasibility.

2 Related Work

A large body of work estimates heterogeneous treatment effects from logged data. The T- and S-learners fit outcome regressions and difference them; the X-learner [5] regresses imputed treatment effects and can exploit treatment imbalance; the R-learner [6] orthogonalizes against a baseline nuisance via Robinson's transformation; and the DR-learner [7, 8] uses the augmented inverse-propensity weighted pseudo-outcome and inherits doubly-robust consistency. Neyman orthogonality reduces first-order sensitivity to nuisance estimation [9]; Assumption 4 specifies the asymptotic nuisance product-rate condition used for our offline learner. Policy learning from doubly-robust scores [10] is the closest precedent for our offline warm start. We use action-specific DR pseudo-outcomes to initialize the CATE estimates from non-uniformly logged data. The online action-centered update then learns the same treatment contrasts from sequential interaction.

LinUCB [1, 11], linear Thompson sampling [2], and regression-oracle contextual bandits [12] target conditional reward. Under our decomposition $\mathbb{E}[Y \mid X, A = a] = f(X) + \tau_a(X)$, that population objective has the same optimizer as incremental value (Eq. (1)); the difference is that a standard linear-reward realization must represent the nuisance f, whereas the semiparametric model need only realize the contrasts. Greenewald et al. [3] introduced action-centered contextual bandits, separating a complex baseline reward from a simple treatment effect and using action-centering to learn the latter without online inverse-propensity weighting. Krishnamurthy et al. [4] developed BOSE for action-independent confounding and obtained the $\widetilde{O}(d\sqrt{T})$ rate without modeling f; its confidence analysis is the basis of our regret argument.

Online resource allocation under budgets is formalized as bandits with knapsacks [13], with contextual algorithms based on policy optimization [14] and adaptive linear programming [15]. A primal-dual approach uses online convex optimization over a dual variable atop a per-step budget-adjusted argmax, with adaptive pacing [16]; recent work refines the contextual frontier [17–19]. These results allow generic reward signals but do not combine a BOSE-style arbitrary action-independent nuisance with an offline causal warm start and a hard spending constraint.

A separate offline line targets cost-aware uplift directly: Zhao and Harinen [20] extend meta-learners to optimize net incremental value across treatments with different costs. This motivates comparison with a static allocation policy. Our Offline-DR baseline fixes the offline treatment-effect estimates and a budget-tuned dual (Section 6.2), while DR-CATE-CBwK updates both online.

Closest in application are recent online causal and budgeted promotion systems. Sawant et al. [21] combine uplift targeting with contextual Thompson sampling but without a budget or regret analysis, and DISCO [22] allocates discounts under a global operational constraint with a live A/B test but targets a reward model rather than the CATE and

gives no regret guarantee. Relative to these systems we add an offline causal warm start, remove an arbitrary shared baseline through action-centering, handle multiple priced actions, and prove a hard-budget regret bound. This coupling provides a semiparametric offline-to-online guarantee for budgeted promotion targeting. On the design side, Kim et al. [23] study experimental-design semiparametric bandits, relevant to our active-set coverage rule.

Closest to our setting, Kim et al. [24] develop doubly-robust Thompson sampling for linear contextual bandits, applying the correction to the raw reward rather than an incremental contrast, and without a budget. Dimakopoulou et al. [25] incorporate causal balancing methods into reward estimation for linear contextual bandits, and Bibaut et al. [26], Hadad et al. [27], Zhan et al. [28] study valid inference after adaptive data collection, motivating our explicit propensity bookkeeping. Our online update builds on action-centered and semiparametric bandits [3, 4]. The distinction is the joint guarantee with an offline doubly robust warm start and a primal-dual hard-budget controller.

3 Problem Setting

3.1 CATE Under Potential Outcomes

For decision epoch $t = 1, \dots, T$: $X_t \in \mathcal{X} \subseteq \mathbb{R}^d$ is the observed context; $A_t \in \mathcal{A} := \{a_0, a_1, \dots, a_K\}$ is the action, with a_0 the no-promotion control; $Y_t \in \mathbb{R}$ is the observed outcome; $c : \mathcal{X} \times \mathcal{A} \to \mathbb{R}_{\geq 0}$ is a known per-decision cost with $c(\cdot, a_0) \equiv 0$; and $B \in \mathbb{R}_{>0}$ is the budget over the horizon, with per-period rate $\rho := B/T$.

Assumption 1 (i.i.d. contexts)**.** Contexts X_t are drawn i.i.d. from a fixed distribution P_X.

Assumption 2 (Structured outcomes and bounded costs)**.** The outcome decomposes as $Y_t = f(X_t) + \tau_{A_t}(X_t) + \xi_t$, where $f(x) = \mathbb{E}[Y \mid X = x, A = a_0]$ is an action-independent baseline with $|f(x)| \leq F$, the treatment contrasts τ_a are the CATEs defined below, and the bounded noise ξ_t is mean-zero and independent of the action given the context and history (Assumption 6). Costs satisfy $c(x, a) \in [0, c_{\max}]$ with $c(x, a) \geq c_{\min} > 0$ for $a \neq a_0$.

Assumption 3 (Bounded contexts and parameters)**.** $\|X_t\|_2 \leq L$ almost surely, and the stacked treatment parameter satisfies $\|\theta^\star\|_2 \leq S$.

We adopt the Neyman-Rubin potential-outcomes framework, in which unit t has potential outcomes $\{Y_t(a) : a \in \mathcal{A}\}$ and only $Y_t = Y_t(A_t)$ is observed. The conditional average treatment effect of action a against the control is

$$\tau_a(x) := \mathbb{E}[\, Y(a) - Y(a_0) \,|\, X = x \,].$$

Under Assumptions 3 and 5, $\tau_a(x) = x^\top \theta_a^\star \in [-R_{\max}, R_{\max}]$ with $R_{\max} := LS$. Identification of τ_a and unbiased off-policy evaluation rely on consistency and no interference (SUTVA), overlap, and unconfoundedness. For the offline log, overlap means $e(a \mid x) := \mathbb{P}(A = a \mid X = x) \geq \epsilon_{\text{off}} > 0$ for every action on the logged context support.

The two phases invoke unconfoundedness differently. Offline, unconfoundedness given the recorded context, $\{Y_t(a)\}_{a \in \mathcal{A}} \perp A_t \mid X_t$, is a maintained assumption, as in observational policy learning [10]. Online it holds by design: with history $\mathcal{H}_{t-1} := \{(X_s, A_s, Y_s)\}_{s<t}$, A_t is drawn from the algorithm's known law π_t, so $\{Y_t(a)\}_{a \in \mathcal{A}} \perp A_t \mid X_t, \mathcal{H}_{t-1}$ and the propensity $P_t = \pi_t(A_t \mid X_t, \mathcal{H}_{t-1})$ is known by construction.

Assumption 4 (Product-of-rates on nuisances)**.** Let $\mu(x, a) := \mathbb{E}[Y \mid X = x, A = a]$ and $e(a \mid x) := \mathbb{P}(A = a \mid X = x)$, and let $\hat{\mu}, \hat{e}$ be cross-fitted estimators over an offline sample of size n_{off}. Then $\|\hat{\mu} - \mu\|_{L^2(P_X)} \, \|\hat{e} - e\|_{L^2(P_X)} = o(n_{\text{off}}^{-1/2})$.

The regret analysis assumes a linearly realizable CATE; nonlinear effects are evaluated empirically in Section 6.7.

Assumption 5 (Linear realizability of the CATE)**.** There exists $\theta_a^\star \in \mathbb{R}^d$ with $\tau_a(x) = x^\top \theta_a^\star$ for all $a \in \mathcal{A}$ and all x in the support of P_X.

3.2 Contextual Bandit With Budget and Regret

A policy $\pi : \mathcal{X} \to \Delta(\mathcal{A})$ induces incremental value and cost

$$V(\pi) := \mathbb{E}_X \mathbb{E}_{A \sim \pi(\cdot|X)}[\, \tau_A(X) \,], \quad C(\pi) := \mathbb{E}_X \mathbb{E}_{A \sim \pi(\cdot|X)}[\, c(X, A) \,]. \tag{2}$$

The incremental objective only reparameterizes the reward objective: for every feasible π,

$$\mathbb{E}_{X, A \sim \pi}[Y(A)] = \mathbb{E}_X[f(X)] + V(\pi), \tag{3}$$

so the constant $\mathbb{E}[f(X)]$ changes neither feasibility nor the maximizer, which we learn without requiring f to lie in the online linear class. We target the rate-based program $\max_\pi V(\pi)$ subject to $C(\pi) \leq \rho$, with Lagrangian

$$L(\pi, \lambda) = \mathbb{E}_X \mathbb{E}_{A \sim \pi(\cdot|X)} \big[\, \tau_A(X) - \lambda \, c(X, A) \,\big] + \lambda \rho. \tag{4}$$

At fixed $\lambda \geq 0$ the inner maximizer is pointwise,

$$\pi^\star_\lambda(x) \in \arg\max_{a \in \mathcal{A}} \big(\tau_a(x) - \lambda\, c(x,a) \big), \tag{5}$$

which motivates the online budget-adjusted score. The fluid LP relaxes the pathwise spending constraint to an expected-cost constraint and supplies the benchmark for cumulative pseudo-regret:

$$R(T) := T \cdot \mathrm{OPT}_{\mathrm{LP}}(\rho) - \sum_{t=1}^{T} \tau_{A_t}(X_t), \quad \mathrm{OPT}_{\mathrm{LP}}(\rho) := \max_{\pi:\, C(\pi) \leq \rho} V(\pi).$$

4 Algorithm

DR-CATE-CBwK has an offline stage that produces a warm-start CATE estimate from a cross-fitted doubly robust learner, and an online stage that, each round, draws a randomized budget-aware action, forms an action-centered feature, and updates a single regularized least-squares estimate of the linear CATE together with a primal-dual budget variable. Two sub-procedures are invoked by the main loop: the offline warm start (Algorithm 2) and the randomized budget-aware action (Algorithm 3).

4.1 Offline Warm Start

We use a doubly robust learner offline to initialize the online estimator. On an offline dataset split into K_{fold} folds (with $k(i)$ the fold of observation i) and nuisances $\hat{\mu}^{(-k)}, \hat{e}^{(-k)}$ trained outside fold k, the cross-fitted DR pseudo-outcome for action a against the control is

$$\begin{aligned} \psi_i(a) = {} & \hat{\mu}^{(-k(i))}(X_i, a) - \hat{\mu}^{(-k(i))}(X_i, a_0) \\ & + \tfrac{\mathbf{1}\{A_i = a\}}{\hat{e}^{(-k(i))}(a \mid X_i)} \big(Y_i - \hat{\mu}^{(-k(i))}(X_i, a) \big) \\ & - \tfrac{\mathbf{1}\{A_i = a_0\}}{\hat{e}^{(-k(i))}(a_0 \mid X_i)} \big(Y_i - \hat{\mu}^{(-k(i))}(X_i, a_0) \big). \end{aligned} \tag{6}$$

When the logging propensities P_i are recorded, as in our setting, $\hat{e}$ is set to the known logging law rather than estimated; otherwise it is cross-fitted, subject to the product-rate condition in Assumption 4. Regressing $\psi_i(a)$ on X_i and stacking the arms gives a warm-start mean $\hat{\theta}^{(0)}$ that seeds the online estimator's prior mean; since $V_0 = \alpha I$, the offline sample initializes the mean while the online design starts isotropically. The outcome model also supplies the clipped baseline $\hat{f}(x) := \Pi_{[-F,F]} \hat{\mu}(x, a_0)$ (used online to residualize), with $|f - \hat{f}| \leq 2F$ whenever $|f| \leq F$. By Neyman orthogonality the nuisance product enters $\hat{\theta}^{(0)}$'s procedure-specific bounds only at second order (Assumption 4); the online theorem instead accepts any valid prior radius, including the deterministic $2S$.

4.2 Online Action-Centered Update

We write the outcome as

$$Y_t = f(X_t) + \tau_{A_t}(X_t) + \xi_t, \qquad \tau_a(x) = x^\top \theta^\star_a, \tag{7}$$

where $f(x) = \mu(x, a_0)$ is an arbitrary, possibly nonlinear, action-independent baseline and ξ_t is mean-zero and bounded given $(X_t, \mathcal{H}_{t-1})$ and independent of the action A_t (Assumption 6). Stack the per-arm parameters into $\theta^\star = (\theta^\star_{a_1}, \ldots, \theta^\star_{a_K}) \in \mathbb{R}^{Kd}$; the control has $\theta^\star_{a_0} = 0$ and is omitted from the stack. Define the block action feature and its mean under the known online action law π_t,

$$z_t(a) = e_a \otimes X_t \in \mathbb{R}^{Kd}, \qquad \bar{z}_t = \sum_{a \in \mathcal{A}^{\mathrm{aff}}_t} \pi_t(a)\, z_t(a), \tag{8}$$

where e_{a_j} is the jth standard basis vector of $\mathbb{R}^K$ and $e_{a_0} = 0$, so that $z_t(a)^\top \theta^\star = \tau_a(X_t)$. Here $\mathcal{A}^{\mathrm{aff}}_t := \{a \in \mathcal{A} : c(X_t, a) \leq B - S_{t-1}\}$ is the affordable set of actions the residual budget $B - S_{t-1}$ can fund, with $S_{t-1} = \sum_{s<t} c(X_s, A_s)$ the cumulative spend; it always contains the cost-free control a_0. We abbreviate $\pi_t(a) := \pi_t(a \mid X_t, \mathcal{H}_{t-1})$. The online update uses the action-centered pair (u_t, g_t),

$$u_t = z_t(A_t) - \bar{z}_t, \qquad g_t = Y_t - \hat{f}(X_t). \tag{9}$$

Because $\bar{z}_t$ is the exact conditional mean of $z_t(A_t)$ under π_t, we have $\mathbb{E}[u_t \mid X_t, \mathcal{H}_{t-1}] = 0$, so the arbitrary baseline, and any error $f - \hat{f}$ in its estimate, cancels in expectation:

$$\mathbb{E}[u_t\, g_t \mid X_t, \mathcal{H}_{t-1}] = \mathbb{E}[u_t u_t^\top \mid X_t, \mathcal{H}_{t-1}]\, \theta^\star =: M_t\, \theta^\star. \tag{10}$$

Algorithm 1 DR-CATE-CBwK: causal contextual bandit with an action-centered update and primal-dual budget control.

Require: Budget B, horizon T, logged dataset $\mathcal{D}$, dual step η_0, confidence radius β_t (set in Theorem 1).

$(\hat{\theta}^{(0)}, \hat{f}, \lambda_1) \leftarrow$ OFFLINEDR($\mathcal{D}$) ▷ Algorithm 2

$V_0 \leftarrow \alpha I$; $b_0 \leftarrow \alpha\hat{\theta}^{(0)}$; $\hat{\theta}_0 \leftarrow \hat{\theta}^{(0)}$; $S_0 \leftarrow 0$

for $t = 1, \ldots, T$ **do**

  Observe X_t.

  $\mathcal{A}_t^{\text{aff}} \leftarrow \{a \in \mathcal{A} : c(X_t, a) \le B - S_{t-1}\}$ ▷ always contains a_0

  $(\pi_t, A_t) \leftarrow$ RANDOMIZEDACTION$(\hat{\theta}_{t-1}, V_{t-1}, X_t, \lambda_t, \mathcal{A}_t^{\text{aff}}; \beta_t)$

  Observe Y_t; incur $c(X_t, A_t)$; $S_t \leftarrow S_{t-1} + c(X_t, A_t)$.

  $\bar{z}_t \leftarrow \sum_{a \in \mathcal{A}_t^{\text{aff}}} \pi_t(a)\, z_t(a)$; $u_t \leftarrow z_t(A_t) - \bar{z}_t$; $g_t \leftarrow Y_t - \hat{f}(X_t)$.

  $V_t \leftarrow V_{t-1} + u_t u_t^\top$; $b_t \leftarrow b_{t-1} + u_t g_t$; $\hat{\theta}_t \leftarrow \arg\min_{\|\theta\|_2 \le S} \|\theta - V_t^{-1} b_t\|_{V_t}$.

  $\lambda_{t+1} \leftarrow \big[\lambda_t + \eta_t(c(X_t, A_t) - \rho)\big]_+$

end for

Algorithm 2 OFFLINEDR: cross-fitted doubly robust warm start.

Require: Logged $\mathcal{D} = \{(X_i, A_i, P_i, Y_i)\}$, folds K_{fold}, ridge α.

for $k = 1, \ldots, K_{\text{fold}}$ **do** fit $\hat{\mu}^{(-k)}$ (and $\hat{e}^{(-k)}$, unless propensities are logged) outside fold k.

end for

Compute $\psi_i(a)$ for each i, a via Eq. (6).

Regress $\psi_i(a)$ on X_i to get the prior mean $\hat{\theta}^{(0)}$; set baseline $\hat{f} \leftarrow \Pi_{[-F,F]}\hat{\mu}(\cdot, a_0)$; project $\hat{\theta}^{(0)}$ onto the radius-S ball; init $\lambda_1 \in [0, R_{\max}/c_{\min}]$ by the offline Lagrangian.

return $(\hat{\theta}^{(0)}, \hat{f}, \lambda_1)$.

This action-centering identity [3, 4] targets the contrasts with no inverse-propensity weight; the score $u_t g_t$ is unbiased for any predictable $\hat{f}$, including $\hat{f} \equiv 0$, so the baseline affects only the variance (Theorem 2(c)). Clipping $\hat{f}$ to $[-F, F]$ gives $\|u_t g_t\|_2 \le 2L(2F + LS + \Xi)$. We maintain the regularized least-squares estimate

$$V_t = V_{t-1} + u_t u_t^\top, \qquad \tilde{\theta}_t = V_t^{-1}\Big(\alpha\hat{\theta}^{(0)} + \textstyle\sum_{s \le t} u_s g_s\Big), \tag{11}$$

with $V_0 = \alpha I$, warm-started at $\hat{\theta}^{(0)}$. Before scoring, we set $\hat{\theta}_t$ to the projection of $\tilde{\theta}_t$ onto $\{\theta : \|\theta\|_2 \le S\}$ in the V_t norm, which cannot enlarge the V_t-error since $\theta^\star$ lies in the set. Because u_t depends on the sampled action, the confidence set is the semiparametric self-normalized bound of Section 5.

4.3 The Action-Centered Estimator and Information Accumulation

The estimator in Eq. (11) accumulates information only through the centered features u_t. At a fixed context the single-round information $M_t = \mathrm{Cov}(e_{A_t}) \otimes X_t X_t^\top$ has rank at most K, hence is singular in $\mathbb{R}^{Kd}$ for $d > 1$, so information must accumulate across contexts. Algorithm 3 is therefore an active-set semiparametric rule [4]: it eliminates every action beaten beyond its joint confidence width, leaving the plausible set $\mathcal{B}_t$ of Eq. (12), and samples within $\mathcal{B}_t$ to meet the optimal-design condition Eq. (14). Imposing that condition only on the active set, not as a uniform floor, lets $\mathcal{B}_t$ collapse to a singleton once a budget-adjusted gap resolves, stopping exploration with no linear-regret term. The farthest-pair rule examines $O(|\mathcal{B}_t|^2)$ pairs and attains $\kappa_0 = 1/4$: if $\Delta_t = z_t(a_t^+) - z_t(a_t^-)$, each centered endpoint is $\pm\Delta_t/2$, so the expected squared centered norm is $\|\Delta_t\|^2_{V_{t-1}^{-1}}/4$, one quarter of the maximum pairwise distance squared.

A mixture assigning weight $\gamma > 0$ to this rule has covariance at least γ times the farthest-pair covariance and hence coverage $\kappa_0 = \gamma/4$. The practical Thompson-restricted alternative need not satisfy this condition. All three rules use the same estimator and affordable set; only rules satisfying coverage with the prescribed confidence parameters carry the regret guarantee.

4.4 Primal-Dual Budget Control

To adapt the budget price as online estimates improve, we use the nonnegative virtual-queue update

$$\lambda_{t+1} := \big[\lambda_t + \eta_t(c(X_t, A_t) - \rho)\big]_+, \tag{13}$$

with step $\eta_t = \eta_0/\sqrt{T}$. Projection only at zero gives a prefix-spend inequality by telescoping the queue. The active-set rule also bounds λ_t: once the price exceeds every plausible treatment gain, only the free control remains active and

Algorithm 3 RANDOMIZEDACTION: active-set semiparametric selection with a budget offset (Assumption 8).

Require: Estimate $\hat{\theta}$, precision V, context X_t, dual λ_t, affordable set $\mathcal{A}_t^{\text{aff}}$; confidence radius β_t (Theorem 1).

1: $\hat{s}_t(a) \leftarrow z_t(a)^\top \hat{\theta} - \lambda_t c(X_t, a)$ for $a \in \mathcal{A}_t^{\text{aff}}$.

2: Form the active set by pairwise elimination,

$$\mathcal{B}_t = \big\{ a \in \mathcal{A}_t^{\text{aff}} : \ \hat{s}_t(a') - \hat{s}_t(a) \le 2\beta_t \|z_t(a') - z_t(a)\|_{V^{-1}} \ \ \forall a' \in \mathcal{A}_t^{\text{aff}} \big\}. \tag{12}$$

3: If $|\mathcal{B}_t| = 1$, put unit mass on its action. Otherwise choose $(a_t^+, a_t^-) \in \arg\max_{a,a' \in \mathcal{B}_t,\, a \neq a'} \|z_t(a) - z_t(a')\|_{V^{-1}}$ and put probability $1/2$ on each endpoint.

4: Sample $A_t \sim \pi_t$; record π_t.

5: **return** (π_t, A_t).

the queue decreases. Thus the update achieves $\widetilde{O}(\sqrt{T})$ dual regret under the theorem's conditions while responding to deviations from the target spending rate ρ. The affordable-set restriction $\mathcal{A}_t^{\text{aff}}$ confines each round's sampling to actions the residual budget can fund and always retains the cost-free control arm, so spending falls back to a_0 once no priced action is affordable; by induction, $S_t \le S_{t-1} + (B - S_{t-1}) = B$ for every round. This pathwise guarantee does not depend on the confidence calibration or coverage condition. An optional uniform exploration floor plays $\pi_t^{\text{floor}} = (1 - \epsilon_{\text{floor}})\pi_t + \epsilon_{\text{floor}} \text{Unif}(\mathcal{A}_t^{\text{aff}})$. Theorem 1 analyzes the default setting $\epsilon_{\text{floor}} = 0$. A positive floor preserves pathwise affordability but changes the queue and regret analysis.

5 Theoretical Analysis

For the coverage-guaranteed instantiation of Algorithm 1, regret decomposes into online bandit regret, dual regret, and a warm-start term. In addition to Assumptions 1, 2, 3, and 5, the analysis uses bounded action-independent noise, active-set coverage, and an offline confidence condition. Context diversity is used separately in the identification discussion of Theorem 2. Throughout, K is the number of priced arms, so the stacked parameter $\theta^\star \in \mathbb{R}^{Kd}$ has effective dimension Kd.

Assumption 6 (Bounded action-independent noise)**.** Conditional on $(X_t, \mathcal{H}_{t-1})$, the noise ξ_t is mean-zero, satisfies $|\xi_t| \le \Xi$ almost surely, and is independent of the sampled action A_t. In particular it is conditionally Ξ-sub-Gaussian.

Unbiasedness of the action-centered score follows from the construction and the known online law (Eq. (10)) for any predictable $\hat{f}$. Since u_t depends on the sampled action, we use the semiparametric self-normalized inequality of Krishnamurthy et al. [4]. It requires the scalar residual to be conditionally independent of u_t given $(X_t, \mathcal{H}_{t-1})$, and Assumption 6 secures exactly this: the residual $\zeta_t = f(X_t) - \hat{f}(X_t) + \bar{z}_t^\top \theta^\star + \xi_t$ has a predictable non-noise part and an action-independent noise ξ_t, so ζ_t is conditionally independent of u_t given $(X_t, \mathcal{H}_{t-1})$. No inverse-propensity weight enters, and clipping the baseline to its known range supplies the bounded residual it needs. The regret proof adopts this action-independent-noise condition of BOSE; the centered moment remains unbiased under action-dependent conditionally mean-zero noise, such as the action-dependent variance of a Bernoulli conversion.

Assumption 7 (Context diversity)**.** There is a constant $\sigma_X^2 > 0$ with $\mathbb{E}[X_t X_t^\top \mid \mathcal{H}_{t-1}] \succeq \sigma_X^2 I$ for all t.

Assumption 8 (Active-set semiparametric selection)**.** With the confidence-based active set $\mathcal{B}_t$ of Eq. (12), the rule of Algorithm 3 draws a predictable distribution π_t supported on $\mathcal{B}_t$ that satisfies the semiparametric optimal-design condition: for a constant $\kappa_0 > 0$ and every pair $a, a' \in \mathcal{B}_t$,

$$\|z_t(a) - z_t(a')\|_{V_{t-1}^{-1}}^2 \le \kappa_0^{-1}\, \mathbb{E}_{A \sim \pi_t}\big[\|z_t(A) - \bar{z}_t\|_{V_{t-1}^{-1}}^2 \,\big|\, X_t, \mathcal{H}_{t-1} \big]. \tag{14}$$

Since $\text{rank}(M_t) \le K$, the single-round information matrix is singular for $d > 1$. Coverage instead controls information in active-set contrast directions. The condition is imposed only over $\mathcal{B}_t$: a blanket floor would explore every round, costing $\Omega(T)$ regret under a persistent gap, whereas $\mathcal{B}_t$ collapses to a singleton once the gap resolves and Eq. (14) becomes automatic. Coverage is supplied by the action-selection rule: the farthest-pair rule attains $\kappa_0 = 1/4$ (Section 4), and Theorem 1 holds for any admissible rule meeting it. The regret bound itself does not use context diversity (Assumption 7): the elliptical-potential sum is algebraic and the coverage condition supplies the per-round exploration. Context diversity enters only the identification discussion of Theorem 2(a), through persistent excitation of every direction.

Assumption 9 (Offline confidence input)**.** Independently of the online interaction, the algorithm is supplied a radius $0 \le \bar{\eta} \le 2S$ such that $\mathbb{P}(\|\hat{\theta}^{(0)} - \theta^\star\|_2 \le \bar{\eta}) \ge 1 - \delta/5$. After projecting $\hat{\theta}^{(0)}$ onto the radius-S ball, the deterministic

choice $\bar{\eta} = 2S$ always satisfies this assumption by the triangle inequality. Any supplied radius can therefore be capped at $2S$.

Optionally, any external finite-sample DR procedure supplying a valid radius η_{off} can be plugged in as $\bar{\eta} = \eta_{\text{off}}$; a root-$n_{\text{off}}$ form $\eta_{\text{off}} = \widetilde{O}(\sqrt{Kd/n_{\text{off}}})$ then yields the warm-start rate below. Such a radius needs procedure-specific overlap, tail, nuisance, and Gram-matrix conditions in addition to the asymptotic product-rate condition in Assumption 4. The generic choice $2S$ provides a valid default.

5.1 Regret Decomposition

Theorem 1 (Regret)**.** *Under Assumptions 1, 2, 3, 5, 6, 8, and 9, let $D := Kd$, set $q_T := 4D\log(9T) + 8\log(20T/\delta)$, $h_T := 27D\log(1 + 2T/D) + 54\log(20T/\delta)$, and choose $\alpha = 4L^2 q_T$. Algorithm 1, run with $\epsilon_{\text{floor}} = 0$ and the BOSE confidence radius*

$$\beta_t \equiv \beta_T := (\Xi + 2F + LS)\sqrt{h_T} + \sqrt{\alpha}\,\bar{\eta}, \tag{15}$$

(where $\bar{\eta}$ is any high-probability upper bound on $\|\hat{\theta}^{(0)} - \theta^\star\|_2$: the generic $\bar{\eta} = 2S$ is always computable, while a certified external radius $\bar{\eta} = \eta_{\text{off}}$ yields the sharper warm-start term below) and step size $\eta_t = \eta_0/\sqrt{T}$ satisfies, for every horizon T, with probability at least $1 - \delta$ over the offline sample and the online interaction,

$$R(T) \le \underbrace{C_1\, D\sqrt{T}\,\log\tfrac{KT}{\delta}}_{R_{\text{online}}} + \underbrace{C_2\sqrt{T}}_{R_{\text{dual}}} + \underbrace{C_3\, D\sqrt{T}\,\bar{\eta}\log\tfrac{KT}{\delta}}_{R_{\text{warm}}}, \tag{16}$$

where C_1, C_2, C_3 depend on Ξ, F, $R_{\max}$, $c_{\max}$, $c_{\min}$, ρ, L, S, κ_0, η_0 and $\log(1/\delta)$ but not on T, n_{off}, d, or K. The constants grow as the budget rate ρ or $c_{\min}$ shrink, so the guarantee is for a fixed positive budget rate. Under an external root-n_{off} guarantee and the choice $\bar{\eta} = \eta_{\text{off}}$, $R_{\text{warm}} = \widetilde{O}(D^{3/2}\sqrt{T/n_{\text{off}}})$. With the generic radius the same overall $\widetilde{O}(D\sqrt{T})$ rate holds. A smaller certified radius tightens R_{warm}, with R_{online} and R_{dual} retaining their stated rates.

The bound is a high-probability frequentist guarantee for the realized $\theta^\star$. The primal term is the regret of a semi-parametric contextual bandit with the budget offset $-\lambda_t c$, at the rate $\widetilde{O}(Kd\sqrt{T})$ of Krishnamurthy et al. [4] for effective dimension Kd. The radius β_T comes from the BOSE self-normalized inequality [4] rather than the ordinary predictable-covariate bound, since u_t is realized only with the action. Equation (15) uses the worst-case residual bound $|f - \hat{f}| \le 2F$, valid for every baseline estimate clipped to $[-F, F]$. The effect of baseline accuracy on score variance is characterized in Theorem 2.

Proof. Let $\mathcal{G}_t = \sigma(\mathcal{H}_{t-1}, X_t)$ and $\mathcal{F}_t = \sigma(\mathcal{G}_t, A_t, Y_t)$, with the offline inputs included in $\mathcal{H}_0$. Set $H := \Xi + 2F + LS$. The residual $\zeta_t = g_t - u_t^\top \theta^\star$ obeys $|\zeta_t| \le H$ and is conditionally independent of u_t given $\mathcal{G}_t$, while $\mathbb{E}[u_t \mid \mathcal{G}_t] = 0$. The BOSE self-normalized bound (Lemmas 14, 16, and 17 of Krishnamurthy et al. [4]), applied to $u_t/(2L)$ and ζ_t/H, gives $\|\sum_{s \le t} u_s \zeta_s\|_{V_t^{-1}} \le H\sqrt{h_T}$ simultaneously for $t \le T$ with failure probability at most $\delta/5$. The normal equations and non-expansiveness of the V_t-norm projection imply

$$\|\hat{\theta}_t - \theta^\star\|_{V_t} \le H\sqrt{h_T} + \sqrt{\alpha}\,\|\hat{\theta}^{(0)} - \theta^\star\|_2 \le \beta_T$$

on intersection with the offline-radius event. This also holds at $t = 0$.

Write $c_t := c(X_t, A_t)$, $G := c_{\max} + \rho$, $\eta := \eta_0/\sqrt{T}$, and $\Phi_t(a, \lambda) := \tau_a(X_t) - \lambda(c(X_t, a) - \rho)$. The projected estimate satisfies $|z_t(a)^\top \hat{\theta}_{t-1}| \le LS$. Since $V_{t-1} \succeq \alpha I$, every priced action is eliminated against the free control when $\lambda_t > (LS + 2\beta_T L/\sqrt{\alpha})/c_{\min}$. The queue then decreases; otherwise it can increase by at most ηG. Induction therefore gives

$$0 \le \lambda_t \le \Lambda := \max\left\{\lambda_1, \frac{LS + 2\beta_T L/\sqrt{\alpha}}{c_{\min}} + \eta G\right\}. \tag{17}$$

Under the prescribed parameters, $\beta_T/\sqrt{\alpha} = O(1)$, so the queue bound is uniform in the horizon.

For each predictable $\lambda_t \ge 0$, weak duality gives $\mathrm{OPT}_{\mathrm{LP}}(\rho) \le \mathbb{E}_X[\max_a \Phi(a, \lambda_t; X)]$. The difference between this conditional expectation and $\max_a \Phi_t(a, \lambda_t)$ is a martingale difference, bounded by $2(R_{\max} + \Lambda G)$. Conditional on the

offline inputs, the queue bound holds on every path. Azuma-Hoeffding therefore gives, with probability at least $1-\delta/5$,

$$\begin{aligned} R(T) \le & \sum_{t=1}^{T}[\max_{a\in\mathcal{A}} \Phi_t(a,\lambda_t) - \Phi_t(A_t,\lambda_t)] \\ & + \sum_{t=1}^{T} \lambda_t(\rho - c_t) + O(\sqrt{T\log(5/\delta)}). \end{aligned} \tag{18}$$

The dual term is controlled deterministically: squaring $\lambda_{t+1} = [\lambda_t + \eta(c_t - \rho)]_+$ and telescoping yields

$$\sum_{t=1}^{T} \lambda_t(\rho - c_t) \le \frac{\lambda_1^2}{2\eta} + \frac{\eta T G^2}{2} = O(\sqrt{T}). \tag{19}$$

To bound the primal term, let $a_t^\star \in \arg\max_{a\in\mathcal{A}_t^{\mathrm{aff}}} \Phi_t(a,\lambda_t)$. On the confidence event, its estimated score disadvantage against any affordable action is at most one confidence width. It therefore survives the two-width elimination test, so both $a_t^\star$ and A_t belong to $\mathcal{B}_t$. The active-set test bounds their estimated score gap by twice the confidence width; the estimation error contributes one more width. Write $\ell_t := \Phi_t(a_t^\star,\lambda_t) - \Phi_t(A_t,\lambda_t)$. Defining $w_t := \mathbb{E}[\|u_t\|^2_{V_{t-1}^{-1}} \mid X_t, \mathcal{H}_{t-1}]$, coverage gives

$$\begin{aligned} \ell_t &\le 3\beta_T \|z_t(a_t^\star) - z_t(A_t)\|_{V_{t-1}^{-1}} \\ &\le 3\beta_T \sqrt{w_t/\kappa_0}. \end{aligned} \tag{20}$$

For $\Delta_t := \|u_t\|^2_{V_{t-1}^{-1}} \le b := 4L^2/\alpha \le 1$, $\mathbb{E}[\Delta_t \mid \mathcal{G}_t] = w_t$. Since $e^{-x} \le 1 - x/2$ on $[0,1]$,

$$\mathbb{E}[e^{-\Delta_t/b} \mid \mathcal{G}_t] \le 1 - w_t/(2b) \le e^{-w_t/(2b)}.$$

The tower property makes $\exp\{\sum_{t\le s}(w_t/(2b) - \Delta_t/b)\}$ a nonnegative $\mathcal{F}_s$-supermartingale starting at one. Markov's inequality gives, with probability at least $1-\delta/5$,

$$\sum_{t=1}^{T} w_t \le 2\sum_{t=1}^{T} \Delta_t + 2b\log(5/\delta). \tag{21}$$

The determinant identity, $\Delta_t \le 1$, and the trace bound yield $\sum_t \Delta_t \le 2\log(\det V_T/\det V_0) \le 2D\log(1 + 4L^2T/(\alpha D))$ [29]. Thus $\sum_t w_t \le \mathcal{P}_T$, where $\mathcal{P}_T := 4D\log(1 + 4L^2T/(\alpha D)) + 2b\log(5/\delta)$. Equation (20) and Cauchy-Schwarz give

$$\sum_{t=1}^{T} \ell_t \le 3\beta_T \sqrt{T\mathcal{P}_T/\kappa_0}. \tag{22}$$

The stochastic component of β_T contributes $\widetilde{O}(D\sqrt{T})$; the prior component contributes

$$\begin{aligned} R_{\mathrm{warm}} &\le 3\sqrt{\alpha}\,\bar{\eta}\sqrt{T\mathcal{P}_T/\kappa_0} \\ &= \widetilde{O}(D\sqrt{T}\,\bar{\eta}). \end{aligned} \tag{23}$$

The definitions of h_T, q_T, and $\mathcal{P}_T$ give the explicit $D\sqrt{T}\log(KT/\delta)$ factors in Eq. (16). An external $\bar{\eta} = \widetilde{O}(\sqrt{D/n_{\mathrm{off}}})$ gives the stated root-n_{off} warm-start rate.

Finally, the queue update also implies $S_t \le t\rho + (\lambda_{t+1} - \lambda_1)/\eta \le t\rho + \Lambda/\eta$. Thus every action is affordable except possibly during the last $\lceil \Lambda/(\eta\rho) + c_{\max}/\rho \rceil = O(\sqrt{T})$ rounds. The global and affordable maxima of Φ_t differ by at most $R_{\max}$: the free control is always affordable and each priced action has $\tau_a - \lambda_t c \le R_{\max}$. Affordability therefore adds only $O(R_{\max}\sqrt{T})$ to Eq. (18) while enforcing $S_t \le B$ on every path. A union bound over the offline-radius, confidence, context-concentration, and potential-concentration events, each with failure probability at most $\delta/5$, proves Eq. (16) with probability at least $1-\delta$. □

5.2 Baseline Robustness of Action-Centering

The action-centered update targets the treatment contrasts τ_a through the score $s_t^{\hat{f}} := u_t\,(Y_t - \hat{f}(X_t))$. We show this is exact for an arbitrary baseline, characterize the bias of a specified separate-arm linear comparator, and identify when oracle residualization reduces total score variance. Fix a round t and condition on $(X_t, \mathcal{H}_{t-1})$; write $r_t := f(X_t) - \hat{f}(X_t)$ for the predictable baseline residual (so $\hat{f} \equiv 0$ gives $r_t = f(X_t)$, the unresidualized centered score). Write $\tau_a := \tau_a(X_t)$ and $\sigma^2(x,a) := \mathrm{Var}(Y_t \mid X_t = x, A_t = a, \mathcal{H}_{t-1})$; expectations and covariances below condition on $(X_t, \mathcal{H}_{t-1})$ unless specified otherwise.

Theorem 2 (Baseline robustness)**.** *Under Assumptions 1, 2, 3, 5, and 6, the following hold. Only the confidence-event bound in (a) and the regret theorem invoke the action-independence of Assumption 6; the moment identity in (a) and the covariance decomposition in (c) require only conditionally mean-zero bounded noise and hold for a general action-dependent conditional variance $\sigma^2(x, a)$.*

(a) For every predictable $\hat{f}$, $\mathbb{E}[s_t^{\hat{f}} \mid X_t, \mathcal{H}_{t-1}] = M_t\theta^\star$, with no inverse-propensity weighting. For $|\hat{f}| \le F$ and under the confidence conditions of Theorem 1, with probability at least $1-\delta$, $|v_t^\top(\hat{\theta}_t - \theta^\star)| \le \beta_t \|v_t\|_{V_t^{-1}}$ for every direction v_t. Directional accuracy is governed by the confidence width $\beta_t\|v_t\|_{V_t^{-1}}$. Context diversity together with persistent randomization spanning every arm block can supply full-parameter identification; neither is required after elimination resolves a contrast. When the active law is a singleton, $M_t = 0$ and that round adds no contrast information.

(b) For an i.i.d. logged sample, consider the separate-arm population OLS coefficients

$$b_a^{\mathrm{raw}} := \arg\min_b \mathbb{E}[(Y - X^\top b)^2 \mid A = a], G_a := \mathbb{E}[XX^\top \mid A = a] \succ 0. \tag{24}$$

Let $\gamma_a := G_a^{-1}\mathbb{E}[Xf(X) \mid A = a]$ be the coefficient of the arm-specific linear projection of f. Then $b_a^{\mathrm{raw}} = \theta_a^\star + \gamma_a$ and the fitted contrast against control has pointwise bias $x^\top(\gamma_a - \gamma_{a_0})$. Under the usual finite-second-moment conditions, empirical separate-arm OLS converges to these coefficients as every arm count diverges. Thus its asymptotic contrast bias in $L^2(P_X)$ is exactly $\|x^\top(\gamma_a - \gamma_{a_0})\|_{L^2(P_X)}$, which is zero under context-independent assignment but can be nonzero under action-dependent context sampling and a misspecified linear reward model. For comparison, by Eq. (3)*, a correctly specified reward learner targets the same constrained optimum as the CATE learner.*

(c) The score covariance splits as

$$\begin{aligned}\mathrm{Cov}(s_t^{\hat{f}} \mid X_t, \mathcal{H}_{t-1}) = \\ \underbrace{\mathbb{E}[u_t u_t^\top \sigma^2(X_t, A_t)]}_{\textit{irreducible noise}} \\ + \underbrace{r_t^2 M_t + 2r_t\,\mathbb{E}[u_t u_t^\top \tau_{A_t}] + \mathrm{Cov}(u_t\tau_{A_t})}_{\textit{signal design}},\end{aligned} \tag{25}$$

where only the terms carrying r_t depend on $\hat{f}$. Comparing the unresidualized score ($\hat{f} \equiv 0$, $r_t = f(X_t)$) with the oracle-residualized learner ($\hat{f} = f$, $r_t = 0$),

$$\mathrm{tr}\,\mathrm{Cov}(s_t^0) - \mathrm{tr}\,\mathrm{Cov}(s_t^f) = f(X_t)^2\,\mathrm{tr}(M_t) + 2f(X_t)\,\mathbb{E}[\|u_t\|_2^2\,\tau_{A_t}], \tag{26}$$

which is strictly positive whenever $\mathrm{tr}(M_t) > 0$ and $|f(X_t)| > 2\,|\mathbb{E}[\|u_t\|_2^2\tau_{A_t}]|/\mathrm{tr}(M_t)$, in particular whenever the baseline scale exceeds twice the treatment-effect scale $R_{\max}$; when the active set is a singleton, $M_t = 0$ and both score covariances vanish.

Part (a) is the semiparametric action-centering identity [3, 4]: because $\mathbb{E}[u_t \mid X_t, \mathcal{H}_{t-1}] = 0$, the baseline and any error in its estimate cancel in the mean, preserving the exact moment under a nonlinear or misspecified baseline. Part (b) is the complementary calculation obtained by substituting the outcome model into the OLS normal equations. Part (c) follows from total covariance conditional on the action and quantifies the oracle reduction in total score variance: exact residualization removes the $O(f^2)$ inflation once the baseline scale dominates the treatment-effect scale. An estimated residualizer always preserves the estimating moment but can increase or decrease variance depending on its error; clipping supplies the valid $2F$ residual bound in Eq. (15).

6 Experiments

We evaluate DR-CATE-CBwK using real job-promotion features and simulated outcomes, treatment effects, logging actions, and offline-to-online shift. Known simulator CATEs and the fluid LP optimum allow direct regret evaluation without estimating counterfactual outcomes from a held-out observational log. We examine operating-regime value, budget feasibility, finite-horizon scaling, baseline robustness, and nonlinear CATEs.

6.1 Experiment Setup

Contexts are drawn from proprietary standardized promotion-log features ($d = 8$ continuous job, company, and poster attributes). There are five total actions: a no-offer control a_0 and $K = 4$ priced promotions (a free trial and \$50, \$100,

\$200 credits), so the stacked parameter has effective dimension $Kd = 32$. The effective per-decision costs are \$15, \$30, \$55, and \$90, respectively, incorporating empirical redemption rates and normalized by the largest cost. The budget is enforced on this normalized effective spend, a proxy for cash liability; a guaranteed cash cap requires a conservative cost bound for each promotion. We impose the linear CATE $\tau_a(x) = x^\top \theta_a^\star$ (Assumption 5) over an action-independent baseline f. Outcomes follow $Y = f(X) + \tau_A(X) + \xi$, with bounded noise $\xi \sim \mathrm{clip}(\mathcal{N}(0, \sigma^2))$ (Assumption 6); unless noted, f is nonlinear, using a tanh-and-interaction surface. The offline log of $n_{\mathrm{off}} = 10^4$ rows is generated by a context-dependent ϵ-greedy logging policy ($\epsilon = 0.15$) on a noisy budget-adjusted CATE proxy, so the logged action depends on the context (the action-dependent-assignment regime of Theorem 2(b), consistent with conditional unconfoundedness); its exact propensities are recorded. An offline-to-online shift parameter in $[0, 1]$ makes the offline log's arm responsiveness differ from the online one; the online environment uses the current, fixed CATE $\theta^\star$, against which $\mathrm{OPT}_{\mathrm{LP}}(\rho)$ and regret are computed exactly. The online horizon is $T = 6000$ at budget rate $\rho = 0.10$; we report means over 32 seeds with 95% confidence intervals, and assess close comparisons with paired-seed bootstrap tests.

The offline warm start is Algorithm 2: gradient-boosted cross-fit nuisances form the DR pseudo-outcome, a ridge regression of which yields $\hat{\theta}^{(0)}$, and $\hat{f} = \hat{\mu}(\cdot, a_0)$ is the baseline model; the online estimator seeds its prior mean at $\hat{\theta}^{(0)}$ with $V_0 = \alpha I$. The online action rule is the practical instantiation of Section 4: active-set elimination (Eq. (12)) followed by Thompson-type randomization restricted to $\mathcal{B}_t$, with the virtual-queue dual update (Eq. (13)) at step $\eta_t = \eta_0/\sqrt{T}$. We form π_t as the explicit empirical categorical law over $\mathcal{B}_t$ induced by a fixed set of posterior draws, so the sampling law, and hence $\bar{z}_t$, is known exactly. We compare this empirical Thompson-restricted rule with farthest-pair and coverage-mixture laws. The latter satisfy Assumption 8, whereas Thompson sampling need not. Theorem 1 analyzes coverage-based selection with prescribed confidence parameters; the experiments use calibrated confidence and dual scales. The affordable-set cap is unchanged across variants.

6.2 Baselines and Ablations

LinUCB [1] fits linear raw rewards, with an affordable-set cap for feasibility. UCB-ALP [15] combines per-arm reward UCBs with fluid-LP pacing. SquareCBwK [17] uses a reward-regression oracle, inverse-gap exploration, and primal-dual control.

Raw-CBwK uses our controller and affordable set with separate-arm linear reward regression, targeting $Q_a - Q_{a_0}$ where $Q_a(x) = \mathbb{E}[Y \mid X = x, A = a]$. Raw-Flex-CBwK additionally uses our offline baseline $\hat{f}$, fitting separate-arm ridge heads to $Y - \hat{f}$; this matches baseline-model capacity but retains uncentered regression. Offline-DR freezes the offline policy with a budget-tuned dual and no online learning.

The No CATE prior ablation sets $\hat{\theta}^{(0)} = 0$ while retaining $\hat{f}$; No residualization sets $\hat{f} = 0$ while retaining the CATE prior. These isolate the prior and residualization components, respectively. The former is the closest BOSE-style learner under our controller [4]. No budget removes both the dual update and the affordable-set restriction.

6.3 Operating Regime: Value and Regret

The operating regime combines a nonlinear baseline, context-dependent logging, and an offline-to-online shift (shift $= 1$). Table 1 reports regret against the fluid LP and total spend. Among all budget-feasible methods DR-CATE-CBwK has the lowest regret (498), ahead of the zero-CATE-prior variant, the capacity-matched Raw-Flex-CBwK, the CBwK and reward baselines, and the frozen offline policy whose stale ranking mis-targets the drifted effects. Paired-seed bootstrap comparisons over the 32 shared seeds show that the CATE prior lowers regret by 94 (95% CI $[29, 160]$, p=0.006) over the zero-CATE-prior variant (with offline residualization retained); residualization by 338 ($[237, 445]$, p<0.001); and DR-CATE-CBwK improves by 220 over Raw-CBwK and 430 over SquareCBwK (both p<0.001). Conditional reward and CATE share the same constrained optimizer (Eq. (3)). A correctly specified reward learner makes the same budget-adjusted comparison, since the shared $f(X_t)$ cancels from $Q_a - \lambda c$. The gap instead reflects these learners' exposure to shared-baseline misspecification and to action-dependent context distributions across separate arm models, together with differences in exploration and budget control. The capacity-matched Raw-Flex-CBwK improves on Raw-CBwK by 98 ($[54, 144]$, p<0.001), a baseline-capacity gap, yet DR-CATE-CBwK still beats it by 122 ($[60, 178]$, p<0.001), so action-centering's robustness survives capacity matching. The only method with lower regret, the no-budget ablation, reaches it only by spending 2729 against the 600 cap, about 4.5 times the budget. The affordable-set restriction enforces feasibility. With the same seeds we evaluate the farthest-pair law and the coverage mixture $\pi_t = (1 - \gamma) q_t^{\mathrm{TS}} + \gamma \pi_t^{\mathrm{FP}}$ (γ=0.1, $\kappa_0 = \gamma/4$; Section 4). Their regrets are 1285 and 822, respectively, versus 498 for Thompson, and both respect the budget. The mixture improves on farthest-pair while retaining active-set coverage, and Thompson achieves the lowest regret in this comparison. DR-CATE-CBwK spends 599.9 of the 600 cap. Its edge over the frozen offline policy is driven by the shift: as the shift sweeps $[0, 1]$ the static policy's regret rises from

257 to approximately 1200 while ours stays near 500 (crossing near 0.4), and Raw-CBwK is flat but uniformly worse under context-dependent logging.

Table 1: Operating regime (nonlinear baseline, context-dependent logging, offline-to-online shift; T=6000, ρ=0.10, 32 seeds, 95% CI). Regret is against the fluid LP (lower is better); $\pm$ is the 95% CI half-width over seeds. DR-CATE-CBwK variants use calibrated parameters as described in Section 6.1. Spend is rounded to whole normalized effective-cost units. Budget-aware methods respect the 600 cap; the No budget ablation spends 2729.

Method	Regret $R(T)$	Spend (B=600)
DR-CATE-CBwK (Thompson, ours)	**498**$_{\pm 56}$	600
No CATE prior	592$_{\pm 54}$	600
Raw-Flex-CBwK	620$_{\pm 38}$	600
Raw-CBwK	718$_{\pm 54}$	600
No residualization	836$_{\pm 98}$	600
SquareCBwK	928$_{\pm 103}$	600
UCB-ALP	998$_{\pm 80}$	594
LinUCB	1040$_{\pm 77}$	600
Offline-DR (fixed)	1214$_{\pm 128}$	582
DR-CATE-CBwK (mixture)	822$_{\pm 48}$	600
DR-CATE-CBwK (farthest-pair)	1285$_{\pm 87}$	600
No budget (infeasible)	$-130_{\pm 91}$	2729

6.4 Regret Rate

To examine finite-horizon scaling, we remove the offline-to-online shift, relax the budget, and use a linear baseline with the fitted offline $\hat{f}$. Figure 1 plots $\log R(T)$ against $\log T$. The practical Thompson rule, labeled "Thompson (deployed)" in the figure, has a fitted exponent of 0.56 up to 6.4×10^4 rounds (bootstrap CI $[0.51, 0.61]$). The mixture and farthest-pair slopes up to 3.2×10^4 rounds are 0.72 ($[0.69, 0.76]$) and 0.85. These estimates describe finite-horizon scaling under the calibrated settings, with Thompson showing the slowest regret growth over the evaluated ranges.

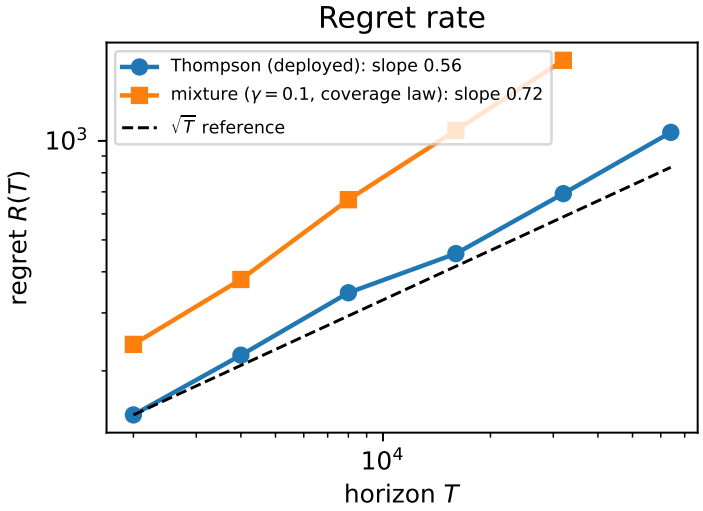


Figure 1: Finite-horizon scaling for Thompson (deployed; slope 0.56, CI $[0.51, 0.61]$) and the coverage mixture (γ=0.1, slope 0.72), using calibrated confidence and dual scales.

6.5 Baseline Robustness

We examine the predictions of Theorem 2 on the estimator, separately from the bandit loop. Figure 2 measures the contrast error $\|\hat{\theta} - \theta^\star\|_2$ as the logged sample grows. Under context-dependent logging the action-centered estimator

(run with $\hat{f} \equiv 0$, so centering alone removes the baseline) has error decreasing throughout, while the separate-arm OLS comparator plateaus at a bias floor, consistent with Theorem 2(a,b) over the evaluated sample sizes. Under randomized logging both estimators' errors decrease, consistent with the role of context-dependent assignment in the comparator's bias. Varying the baseline magnitude at fixed sample size gives approximately constant error for the residualized estimator and linearly increasing error for the unresidualized centered estimator ($\hat{f} \equiv 0$), consistent with Theorem 2(c). Residualizer accuracy also matters: with $|f| \approx 4R_{\max}$ the contrast error is 0.17 under an oracle $\hat{f}$ and 3.2 with none, but a badly misspecified $\hat{f}$ raises it to 8.1, above the unresidualized estimator, illustrating how residualization quality affects variance.

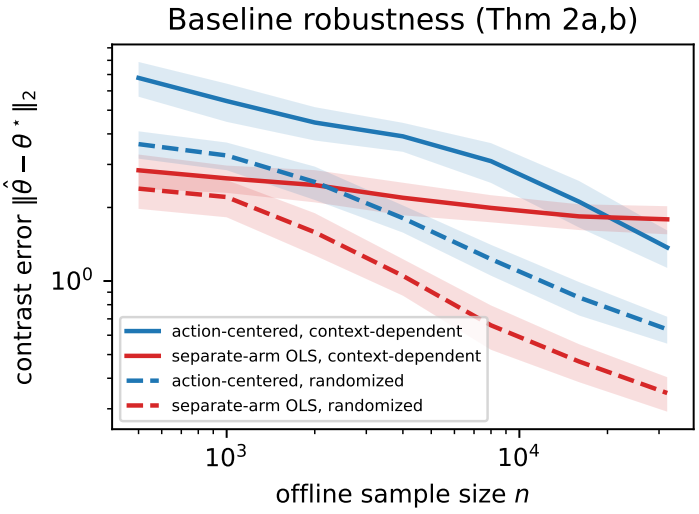


Figure 2: Baseline robustness (Theorem 2(a,b)). Contrast error versus sample size. Under context-dependent logging the action-centered estimator's error keeps decreasing while separate-arm OLS plateaus at a bias; under randomized logging both decrease. Means over 32 seeds.

6.6 Coverage, Sensitivity, and Calibration

We measure the coverage condition for the empirical action rules (Assumption 8) empirically by logging, per multi-action round, the achieved constant $\kappa_t := w_t / \max_{a \neq a' \in \mathcal{B}_t} \|z_t(a) - z_t(a')\|^2_{V_{t-1}^{-1}}$ (zero-distance pairs excluded), where $w_t = \mathbb{E}_{A \sim \pi_t}[\|z_t(A) - \bar{z}_t\|^2_{V_{t-1}^{-1}} \mid X_t, \mathcal{H}_{t-1}]$. This is the largest constant certified by Eq. (14). The farthest-pair rule attains $\kappa_0 = 1/4$ in each evaluated multi-action round. The Thompson rule has median coverage 0 and mean 0.02 over $\approx 4.7 \times 10^3$ such rounds per seed, as it concentrates draws on the apparent best arm. It achieves lower regret (Table 1), illustrating the gap between sufficient coverage and the empirical behavior of Thompson sampling. The affordable-set restriction preserves budget feasibility for both rules.

The raw offline error grows with drift ($\|\hat{\theta}^{(0)}_{\text{raw}} - \theta^\star\| \approx 1.53$ stationary, 2.63 under full drift). With $S = 2$ (Assumption 3; $\|\theta^\star\| \approx 0.8$), projection onto the radius-S ball leaves post-projection errors ≈ 1.5 and 2.0, within $\bar{\eta} = 2S = 4$. The larger initial error under shift motivates online adaptation. The empirical implementation uses calibrated confidence, Thompson covariance, and dual scales rather than the theorem's bound-based settings. Scales are fixed across seeds. A post-hoc 27-point sensitivity grid gives regret in $[345, 773]$ (median 537). Scoring against the known simulator optimum is oracle-assisted; a practical calibration would use an estimable validation objective.

6.7 Robustness to the Linear-CATE Assumption: a Neural Variant

To examine CATE misspecification (Assumption 5), we replace the linear CATE by a nonlinear, non-monotone surface ($\tau_a(x) = \kappa[\sin(\omega\, x^\top u_a) + \sin(\omega\, x^\top v_a + 1)]$, with amplitude κ, frequency $\omega = 5$, and fixed random per-arm directions u_a, v_a) that a linear head cannot represent, keeping everything else fixed. The neural variant (Neural DR-CATE-CBwK) runs the action-centered algorithm in a learned feature space: the offline stage fits a one-hidden-layer $\tanh$ network to

the cross-fitted contrasts and uses its representation $\phi(x)$ in place of x ($e_a \otimes \phi(X_t)$), leaving everything causal and the budget controller unchanged. This provides an empirical extension to learned representations.

Table 2 reports the result. Under a nonlinear CATE the linear DR-CATE-CBwK has regret 945, lower than every reward-based baseline (Table 2), with the margin over the capacity-matched Raw-Flex-CBwK confirmed by a paired test (64, $[28, 100]$, p=0.001). Action-centering still cancels the shared baseline in the moment. The neural variant has regret 906; its improvement over the linear head (39, paired p=0.11) is not statistically significant. Both variants retain affordable-set budget feasibility.

Table 2: Nonlinear CATEs ($\omega = 5$, T=6000, ρ=0.10, 32 seeds; $\pm$ denotes 95% CI half-width). The neural improvement over the linear variant is not significant (p=0.11).

Method	Regret $R(T)$
Neural DR-CATE-CBwK	$\mathbf{906}_{\pm 42}$
DR-CATE-CBwK (linear)	$945_{\pm 45}$
Raw-Flex-CBwK	$1009_{\pm 45}$
Raw-CBwK	$1063_{\pm 46}$
UCB-ALP	$1070_{\pm 50}$
SquareCBwK	$1078_{\pm 57}$
LinUCB	$1088_{\pm 50}$

7 Conclusion

We presented DR-CATE-CBwK, an online causal contextual bandit combining a doubly robust CATE warm start, BOSE-style action-centering, and a global budget. The joint design couples an offline causal prior and semiparametric active-set learning with a primal-dual budget controller. It therefore avoids modeling the arbitrary baseline online and does not require permanent exploration. For linear CATEs in the stated stationary i.i.d. setting, Theorem 1 gives a high-probability $\widetilde{O}(Kd\sqrt{T})$ bound against the fluid LP for the coverage-guaranteed action rule. Theorem 2 establishes baseline-free centered moments, characterizes when a specified separate-arm comparator can be biased, and identifies when oracle residualization reduces score variance. On a semi-synthetic benchmark based on an online job-promotion targeting dataset, the practical Thompson-restricted variant improves on a raw-reward learner and a static offline policy under an offline-to-online shift and shows sublinear finite-horizon scaling, while the separately evaluated farthest-pair and coverage-mixture variants are more conservative. The centered estimator recovers contrasts where the specified uncentered learner is biased, and all budget-aware variants respect the effective-cost budget (Section 6). Action-centering supplies the statistical robustness, the offline prior improves adaptation to an offline-to-online shift, and the affordable-set primal-dual controller supplies pathwise feasibility. Together, these components turn incremental-effect estimation into an online, budget-aware decision process while preserving the distinction between the coverage-guaranteed action laws and the stronger empirical Thompson variant.

Each structural choice suggests a natural extension: continuous doses via kernel smoothing, non-i.i.d. data via discounted updates, and multiple resources via a multi-dimensional dual. The supplement adds frozen-representation and online NTK extensions under explicit representation-quality and fixed-NTK coupling conditions. Our guarantees assume a fixed positive budget rate, motivating further study of residual-budget-adaptive exploration and the optimal small-budget rate. Public uplift datasets and live traffic provide natural next steps for validation.

Ethical Considerations

Promotion targeting allocates financial benefits across users. Equitable allocation requires attention to historical disparities: underserved groups may have noisier effect estimates and receive fewer offers, further reducing opportunities to learn their responses. Deployment should include allocation and outcome audits across protected and vulnerable groups, human oversight, and recourse alongside the financial budget.

Offline causal estimates depend on unconfoundedness given the recorded features, so logging validation should assess potential unobserved confounding. Online randomization supports identification, but eliminated actions may receive no further observations. Randomized holdouts with suitable overlap and monitoring of predicted versus observed effects are needed to detect persistent errors and feedback effects. Valid evaluation of an alternative policy requires overlap between its actions and those represented in the evaluation data.

Behavioral features and propensities require data minimization, access controls, and retention limits. Sensitive attributes should be used only when lawful and necessary, including for audited fairness assessments. Operators should protect logs and online updates against poisoning, feature manipulation, and unauthorized access. Appropriate offer restrictions, exposure limits, and human review are needed to prevent manipulative targeting. The affordable-set rule caps the specified effective costs; a cash guarantee additionally requires conservative cost accounting for variation in redemption.

Our experiments use de-identified logged features and simulated outcomes. For deployment, we recommend causal validation, privacy review, security controls, and assessment of distributional harms.

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Appendix

The appendix contains the action-centered confidence and design results, the budget-rate justification, the proof of Theorem 2, the frozen neural variant, and the online neural-tangent extension with its supporting proofs. The proof of Theorem 1 appears in Section 5.

A Action-Centered Least-Squares Confidence Set

Let $D = Kd$, $\mathcal{G}_t = \sigma(\mathcal{H}_{t-1}, X_t)$ be the pre-action filtration, and $\mathcal{F}_t = \sigma(\mathcal{G}_t, A_t, Y_t)$ the post-outcome filtration, with the offline inputs included in $\mathcal{H}_0$. Take $L > 0$ and a predictable baseline estimate clipped to $[-F, F]$. Writing $\zeta_t := g_t - u_t^\top \theta^\star = f(X_t) - \hat{f}(X_t) + \bar{z}_t^\top \theta^\star + \xi_t$, we have $\|u_t\|_2 \le 2L$, $|\zeta_t| \le H := 2F + LS + \Xi$, and, conditional on $\mathcal{G}_t$, u_t is centered while $\mathbb{E}[\xi_t \mid \mathcal{G}_t, A_t] = 0$. Since the non-noise part of ζ_t is $\mathcal{G}_t$-measurable, $u_t\zeta_t$ is $\mathcal{F}_t$-measurable with conditional mean zero. Moreover, by Assumption 6 the noise ξ_t is independent of A_t given $\mathcal{G}_t$ and the remaining part of ζ_t is $\mathcal{G}_t$-measurable, so ζ_t is conditionally independent of u_t given $\mathcal{G}_t$; this is the hypothesis under which the BOSE self-normalized inequality (Lemmas 14, 16, 17 of Krishnamurthy et al. [4]) controls $\sum_t u_t\zeta_t$. The normalizer nevertheless contains the same non-predictable u_t, so the ordinary predictable-covariate theorem of Abbasi-Yadkori et al. [29] does not apply.

Lemma A.1 (Multi-action BOSE confidence set)**.** *Set $q_T = 4D\log(9T) + 8\log(20T/\delta)$ and $h_T = 27D\log(1 + 2T/D) + 54\log(20T/\delta)$. Under Assumptions 2, 3, 5, and 6, if $\alpha \ge 4L^2 q_T$, then, with probability at least $1 - \delta/5$, simultaneously for $t \le T$,*

$$\|\hat{\theta}_t - \theta^\star\|_{V_t} \le H\sqrt{h_T} + \sqrt{\alpha}\,\|\hat{\theta}^{(0)} - \theta^\star\|_2.$$

Lemma A.1 gives Eq. (15) with any valid $\bar{\eta}$. Since $\theta^\star$ lies in the radius-S ball, projecting $\hat{\theta}^{(0)}$ onto that ball gives the generic $\bar{\eta} = 2S$; an external offline confidence procedure may give a sharper data-dependent choice. Clipping $\hat{f}$ is used only to make $|f - \hat{f}| \le 2F$; action-centering, not clipping, removes its bias.

Lemma A.2 (Multi-action farthest-pair design)**.** *For any nonempty finite active set $\mathcal{B}$, features $z(a)$, and positive definite V, the distribution in Algorithm 3 satisfies, for all $a, a' \in \mathcal{B}$,*

$$\|z(a) - z(a')\|_{V^{-1}}^2 \le 4\,\mathbb{E}_{A\sim\pi}\|z(A) - \bar{z}\|_{V^{-1}}^2.$$

Corollary A.3 (Mixture coverage)**.** *Fix $\gamma \in (0, 1]$ and let $\pi_t = (1-\gamma)q_t^{\mathrm{TS}} + \gamma\pi_t^{\mathrm{FP}}$, where q_t^{TS} is any predictable law on $\mathcal{B}_t$ and π_t^{FP} is the farthest-pair law of Lemma A.2. Then π_t satisfies Eq.* (14) *with $\kappa_0 = \gamma/4$.*

Proof. The round potential $w_t^\mu := \mathbb{E}_{A\sim\mu}\|z_t(A) - \bar{z}_\mu\|_{V_{t-1}^{-1}}^2$ is the V_{t-1}^{-1}-trace of $\mathrm{Cov}_\mu(z_t(A))$. For a mixture $\mu = (1-\gamma)q + \gamma r$,

$$\begin{aligned}\mathrm{Cov}_\mu(z) &= (1-\gamma)\mathrm{Cov}_q(z) + \gamma\,\mathrm{Cov}_r(z)\\ &\quad + \gamma(1-\gamma)(\mu_q - \mu_r)(\mu_q - \mu_r)^\top \succeq \gamma\,\mathrm{Cov}_r(z),\end{aligned}$$

so $w_t^{\mathrm{mix}} \ge \gamma\, w_t^{\mathrm{FP}}$. By Lemma A.2, every pair obeys $\|z_t(a) - z_t(a')\|_{V_{t-1}^{-1}}^2 \le 4w_t^{\mathrm{FP}} \le (4/\gamma)\, w_t^{\mathrm{mix}}$, which is Eq. (14) with $\kappa_0 = \gamma/4$. □

B The Per-Period Budget Rate: Formal Justification

The algorithm enforces the almost-sure cumulative cap $\sum_t c(X_t, A_t) \le B$ through its affordable-action rule. The fluid benchmark relaxes this cap to the expected-cost rate $\rho := B/T$, making the Lagrangian Eq. (4) time-homogeneous and its inner maximization decompose pointwise across contexts as in Eq. (5). We show that a stationary rate-feasible policy can be truncated to satisfy the pathwise cap, the rate program has strong duality, and, for a fixed positive budget rate, the fluid benchmark exceeds the optimal hard-budget value by at most $\widetilde{O}(\sqrt{T})$. Fix T and $B = T\rho$ with $\rho > 0$; write $\mathcal{F}_t := \sigma(X_1, A_1, \ldots, X_t, A_t)$, $S_T := \sum_t c(X_t, A_t)$, $r_{\max} := \sup_{x,a}|\tau_a(x)| < \infty$, and, by Assumption 2, $0 \le c \le c_{\max}$. The rate program is

$$\text{(P-Rate)} \quad \max_{\pi} V(\pi) \text{ s.t. } C(\pi) \le \rho, \tag{B.1}$$

with V, C as in Eq. (2). The cumulative program (P-Cum) maximizes $\mathbb{E}[\sum_{t=1}^T \tau_{A_t}(X_t)]$ over nonanticipating policies subject to the almost-sure cap $S_T \le B$.

Lemma B.1 (Concentration of cumulative spend)**.** *For any (possibly adaptive) policy and any $\delta \in (0, 1)$, with probability at least $1 - \delta$, $\max_{s\le T}\big|S_s - \sum_{t\le s}\mathbb{E}[c(X_t, A_t) \mid \mathcal{F}_{t-1}]\big| \le c_{\max}\sqrt{2T\log(2/\delta)}$.*

Corollary B.2. *If π is stationary and rate-feasible, then with probability at least $1-\delta$, simultaneously for $s \le T$, $S_s \le s\rho + c_{\max}\sqrt{2T\log(2/\delta)}$.*

Proposition B.3 (Truncation gives path-wise feasibility)**.** *Let π be stationary with $C(\pi) \le \rho$. Draw fresh proposals $A_t \sim \pi(\cdot \mid X_t)$ and let $\tilde{\pi}$ follow them until the first proposal whose cost exceeds the remaining budget; from that round onward it plays a_0. Then $\sum_t c(X_t, \tilde{A}_t) \le B$ almost surely and $\left|T\,V(\pi) - \mathbb{E}[\sum_t \tau_{\tilde{A}_t}(X_t)]\right| = O((r_{\max}c_{\max}/\rho)\sqrt{T\log(2T)})$.*

Proposition B.4 (Strong duality of the rate-constrained program)**.** *Under Assumptions 1 and 2, with bounded measurable treatment effects as above, Slater's condition is met by $\pi(a \mid x) = \mathbf{1}\{a = a_0\}$, since $c(\cdot, a_0) \equiv 0$ and $\rho > 0$. Program Eq.* (B.1) *is a convex program over stochastic kernels (equivalently, occupancy measures $\nu(dx, a) = P_X(dx)\pi(a \mid x)$) and strong duality holds: $\mathrm{OPT}_{\mathrm{LP}}(\rho) = \max_{C(\pi)\le\rho} V(\pi) = \min_{\lambda\ge 0}\max_{\pi} L(\pi, \lambda)$. At any optimal $\lambda^\star$ the inner maximizer is pointwise, $\pi^\star_{\lambda^\star}(x) \in \arg\max_{a\in\mathcal{A}}(\tau_a(x) - \lambda^\star c(x, a))$. A primal-optimal kernel is supported on these maximizers almost surely and may need to randomize among tied actions to satisfy the cost constraint.*

Proposition B.5 (Fluid LP versus hard-budget benchmark)**.** *Let $\mathrm{OPT}_{\mathrm{pathwise}}(B)$ be the optimum over all history-dependent policies under the almost-sure cap $S_T \le B$. Under the assumptions of Proposition B.4, $0 \le T\cdot\mathrm{OPT}_{\mathrm{LP}}(\rho) - \mathrm{OPT}_{\mathrm{pathwise}}(B) = O((r_{\max}c_{\max}/\rho)\sqrt{T\log(2T)})$, and hence the gap is $o(T)$ for fixed $r_{\max}$, $c_{\max}$, and $\rho > 0$.*

C Proofs of the Confidence and Design Lemmas

Proof of Lemma A.1. If $H = 0$, then $\zeta_t = 0$ almost surely and the stochastic term below vanishes. Otherwise, the following argument bounds that term. Normalize $Z_t := u_t/(2L)$ and $\tilde{\zeta}_t := \zeta_t/H$, so $\|Z_t\|_2 \le 1$ and $|\tilde{\zeta}_t| \le 1$. Conditional on $\mathcal{G}_t$ we have $\mathbb{E}[u_t \mid \mathcal{G}_t] = 0$, the non-noise part of ζ_t is $\mathcal{G}_t$-measurable, and $\mathbb{E}[\xi_t \mid \mathcal{G}_t, A_t] = 0$; hence $\mathbb{E}[u_t\zeta_t \mid \mathcal{G}_t] = 0$ and $\{u_t\zeta_t\}$ is a martingale-difference sequence adapted to $\{\mathcal{F}_t\}$. Under a general action law, Z_t need not be conditionally symmetric. Moreover, it is realized after action sampling, so the predictable-covariate theorem of Abbasi-Yadkori et al. [29] cannot be applied directly. Let $\widehat{\Sigma}_t := \sum_{s\le t} Z_s Z_s^\top$ and $\Sigma_t := \sum_{s\le t}\mathbb{E}[Z_s Z_s^\top \mid \mathcal{G}_s]$. Set $a_T := 9D\log(9T) + 8\log(20T/\delta)$ and $\lambda_0 := \alpha/(4L^2) \ge q_T$. Lemma 14 of Krishnamurthy et al. [4], applied at each prefix with failure probability $\delta/(10T)$, gives $\Sigma_t \preceq 2\widehat{\Sigma}_t + a_T I$ simultaneously for $t \le T$ except on an event of probability at most $\delta/10$. Define

$$Q_T := 3\lambda_0 - a_T \ge 3D\log(9T) + 16\log(20T/\delta) \ge 1.$$

On this covariance event,

$$\frac{V_t}{4L^2} = \lambda_0 I + \widehat{\Sigma}_t \succeq \frac{Q_T I + \widehat{\Sigma}_t + \Sigma_t}{3}.$$

Independently of conditioning on that event, their Lemma 16 applies to $(Z_s, \tilde{\zeta}_s)$ with ridge $Q_T I$ and unit residual bound. A union bound with failure probability $\delta/(10T)$ at each prefix, together with $\mathrm{tr}(\widehat{\Sigma}_t + \Sigma_t) \le 2T$ and $Q_T \ge 1$, gives

$$\left\|\sum_{s\le t} Z_s\tilde{\zeta}_s\right\|^2_{(Q_T I + \widehat{\Sigma}_t + \Sigma_t)^{-1}} \le D\log(1 + 2T/D) + 2\log(10T/\delta)$$

simultaneously except on an event of probability at most $\delta/10$. Intersecting the two events and rescaling therefore yields

$$\left\|\sum_{s\le t} u_s\zeta_s\right\|^2_{V_t^{-1}} \le H^2\{3D\log(1 + 2T/D) + 6\log(10T/\delta)\} \le H^2 h_T$$

with total failure probability at most $\delta/5$. This is the empirical normalizer argument underlying their Lemma 17. It holds for the adaptive, $\mathcal{G}_t$-measurable affordable set, active set, and law π_t. For the unprojected estimate $\tilde{\theta}_t = V_t^{-1}(\alpha\hat{\theta}^{(0)} + \sum_{s\le t} u_s g_s)$ we have $V_t(\tilde{\theta}_t - \theta^\star) = \alpha(\hat{\theta}^{(0)} - \theta^\star) + \sum_{s\le t} u_s\zeta_s$, so by the triangle inequality and $V_t \succeq \alpha I$,

$$\begin{aligned}\|\tilde{\theta}_t - \theta^\star\|_{V_t} &\le \Big\|\sum_{s\le t} u_s\zeta_s\Big\|_{V_t^{-1}} + \alpha\|\hat{\theta}^{(0)} - \theta^\star\|_{V_t^{-1}} \\ &\le H\sqrt{h_T} + \sqrt{\alpha}\,\|\hat{\theta}^{(0)} - \theta^\star\|_2.\end{aligned}$$

The V_t-norm projection onto the radius-S ball, which contains $\theta^\star$, is non-expansive, so the same bound holds for the projected $\hat{\theta}_t$. □

Proof of Lemma A.2. The singleton case is immediate. Otherwise let (a^+, a^-) attain the active-set diameter $d^2 := \max_{a,a' \in \mathcal{B}} \|z(a) - z(a')\|_{V^{-1}}^2$, and let π place mass $1/2$ on each of a^+, a^-. Then $\bar{z} = \frac{1}{2}(z(a^+) + z(a^-))$, so $z(a^\pm) - \bar{z} = \pm\frac{1}{2}(z(a^+) - z(a^-))$ and each endpoint has squared centered norm $d^2/4$; hence $\mathbb{E}_{A\sim\pi}\|z(A) - \bar{z}\|_{V^{-1}}^2 = d^2/4$. Every pair $a, a' \in \mathcal{B}$ obeys $\|z(a) - z(a')\|_{V^{-1}}^2 \le d^2 = 4\,\mathbb{E}_{A\sim\pi}\|z(A) - \bar{z}\|_{V^{-1}}^2$, which is Eq. (14) with $\kappa_0 = 1/4$, independent of $|\mathcal{B}|$. □

D Proofs for the Per-Period Budget Rate

Proof of Lemma B.1. The differences $c(X_t, A_t) - \mathbb{E}[c(X_t, A_t) \mid \mathcal{F}_{t-1}]$ form a martingale-difference sequence bounded in absolute value by $c_{\max}$ (Assumption 2). Applying the maximal Azuma-Hoeffding inequality to the partial sums $S_s - \sum_{t\le s} \mathbb{E}[c(X_t, A_t) \mid \mathcal{F}_{t-1}]$ gives, with probability at least $1 - \delta$, the uniform (in $s \le T$) deviation bound $c_{\max}\sqrt{2T\log(2/\delta)}$. □

Proof of Corollary B.2. A stationary rate-feasible π has $\mathbb{E}[c(X_t, A_t) \mid \mathcal{F}_{t-1}] = C(\pi) \le \rho$, so $\sum_{t\le s} \mathbb{E}[c(X_t, A_t) \mid \mathcal{F}_{t-1}] \le s\rho$; substituting into Lemma B.1 gives $S_s \le s\rho + c_{\max}\sqrt{2T\log(2/\delta)}$ simultaneously for $s \le T$. □

Proof of Proposition B.3. Couple $\tilde{\pi}$ to an untruncated sequence of proposals from π, and let S_s denote the spend of that proposal sequence. Take $\delta = (T+1)^{-2}$ in Corollary B.2 and set $D_T := c_{\max}\sqrt{2T\log(2(T+1)^2)}$ and $m := \lceil (D_T + c_{\max})/\rho \rceil$. On the resulting event, if $m < T$, every prefix with $s \le T - m$ satisfies $S_s \le s\rho + D_T \le B - c_{\max}$. Thus no proposal is rejected before the final m rounds. If $m \ge T$, at most $T \le m$ rounds can change. On every path, the actual actions follow affordable proposals until the first rejection and then use the cost-free control, so cumulative spend never exceeds B. Each changed round alters the incremental value by at most $r_{\max}$. The expected reward of the untruncated sequence is $TV(\pi)$, so

$$\left| TV(\pi) - \mathbb{E}\left[\sum_t \tau_{\tilde{A}_t}(X_t)\right] \right| \le r_{\max} m + \frac{r_{\max} T}{(T+1)^2} = O\left(\frac{r_{\max} c_{\max}}{\rho}\sqrt{T\log(2T)}\right).$$

□

Proof of Proposition B.4. Represent a kernel by its finitely many coordinates in $L^\infty(P_X)^{K+1}$. The constraints $\pi_a \ge 0$ and $\sum_a \pi_a = 1$ define a weak-star closed subset of the product unit ball, which is compact by Banach-Alaoglu. Since the costs and treatment effects are bounded and measurable, C and V are weak-star continuous linear functionals. Consequently, the achievable set of pairs $(C(\pi), V(\pi))$ is compact and convex, and the rate-constrained maximum is attained. The strictly feasible control-only kernel has $C = 0 < \rho$. A supporting-hyperplane argument for this two-dimensional achievable set therefore gives strong duality and a finite optimal multiplier:

$$\mathrm{OPT}_{\mathrm{LP}}(\rho) = \min_{\lambda \ge 0} \max_{\pi} L(\pi, \lambda).$$

For fixed λ, finite-action measurable maximization gives

$$\max_{\pi} L(\pi, \lambda) = \lambda\rho + \mathbb{E}_X \max_a \{\tau_a(X) - \lambda c(X, a)\}.$$

Thus a deterministic pointwise maximizer attains the inner maximum. At an optimal multiplier, a primal-optimal kernel assigns mass only to pointwise maximizers and satisfies complementary slackness. Randomization among ties can be necessary for primal feasibility; an arbitrary deterministic tie choice need not satisfy the budget rate. □

Proof of Proposition B.5. For the upper bound, any history-dependent policy feasible for the almost-sure cap $S_T \le B$ satisfies $\mathbb{E}[S_T] \le B$, so its time-averaged occupancy $\bar{\nu} := \frac{1}{T}\sum_t \nu_t$, where $\nu_t(E, a) := \mathbb{P}(X_t \in E, A_t = a)$, has context marginal P_X and is rate-feasible with $V(\bar{\nu}) = \frac{1}{T}\mathbb{E}[\sum_t \tau_{A_t}(X_t)]$; hence $\mathrm{OPT}_{\mathrm{pathwise}}(B) \le T\,\mathrm{OPT}_{\mathrm{LP}}(\rho)$. For the lower bound, applying Proposition B.3 to a fluid-optimal stationary kernel achieves expected cumulative value at least $T\,\mathrm{OPT}_{\mathrm{LP}}(\rho) - O((r_{\max}c_{\max}/\rho)\sqrt{T\log(2T)})$ while satisfying the cap on every path. Combining the bounds proves the claim. □

E Proof of Theorem 2

Proof. Fix a round t and condition on $(X_t, \mathcal{H}_{t-1})$; all expectations below are over $A_t \sim \pi_t(\cdot \mid X_t)$ and the noise ξ_t. Recall $Y_t = f(X_t) + \tau_{A_t}(X_t) + \xi_t$ with $\mathbb{E}[\xi_t \mid X_t, A_t] = 0$ and $\mathrm{Var}(\xi_t \mid X_t, A_t) = \sigma^2(X_t, A_t)$, the centered feature $u_t = z_t(A_t) - \bar{z}_t$ with $\mathbb{E}[u_t \mid X_t, \mathcal{H}_{t-1}] = 0$, and $M_t = \mathbb{E}[u_t u_t^\top]$. Write $r_t = f(X_t) - \hat{f}(X_t)$, a constant given the predictable $\hat{f}$, and $s_t^{\hat{f}} = u_t(Y_t - \hat{f}(X_t))$.

For part (a), since $Y_t - \hat{f}(X_t) = r_t + z_t(A_t)^\top \theta^\star + \xi_t$,

$$\begin{aligned}\mathbb{E}[s_t^{\hat{f}} \mid X_t, \mathcal{H}_{t-1}] \\ = r_t\, \mathbb{E}[u_t] + \mathbb{E}[u_t z_t(A_t)^\top]\theta^\star + \mathbb{E}[u_t \xi_t].\end{aligned}$$

The first term vanishes by $\mathbb{E}[u_t] = 0$; the third vanishes because $\mathbb{E}[\xi_t \mid X_t, A_t] = 0$. For the second, $\mathbb{E}[u_t z_t(A_t)^\top] = \mathbb{E}[u_t(u_t + \bar{z}_t)^\top] = \mathbb{E}[u_t u_t^\top] + \mathbb{E}[u_t]\bar{z}_t^\top = M_t$, so $\mathbb{E}[s_t^{\hat{f}} \mid X_t, \mathcal{H}_{t-1}] = M_t\theta^\star$ for every predictable $\hat{f}$, including $\hat{f} \equiv 0$. The regularized least-squares estimator Eq. (11) solves the empirical version of $M_t\theta = \mathbb{E}[s_t^{\hat{f}}]$. Lemma A.1 gives $|v_t^\top(\hat{\theta}_t - \theta^\star)| \le \beta_t \|v_t\|_{V_t^{-1}}$. Thus every direction satisfying $\beta_t\|v_t\|_{V_t^{-1}} \to 0$ has vanishing estimation error on the confidence event. Persistent exploration must make the width shrink to obtain this conclusion. Directions used only by eliminated actions need not remain excited, and full-parameter consistency is not claimed.

For part (b), with the i.i.d. logged sample of Eq. (24), the population normal equations and $\mathbb{E}[\xi \mid X, A] = 0$ give

$$\begin{aligned}G_a b_a^{\mathrm{raw}} &= \mathbb{E}[XY \mid A = a] \\ &= \mathbb{E}[Xf(X) \mid A = a] + G_a\theta_a^\star.\end{aligned}$$

Since $G_a \succ 0$, it follows that $b_a^{\mathrm{raw}} = \theta_a^\star + \gamma_a$ with $\gamma_a = G_a^{-1}\mathbb{E}[Xf(X) \mid A = a]$. The control block has $\theta_{a_0}^\star = 0$, so the separate-arm fitted contrast at x differs from $\tau_a(x)$ by exactly $x^\top(\gamma_a - \gamma_{a_0})$. The law of large numbers and continuity of matrix inversion give convergence of empirical arm-specific OLS to these coefficients whenever every arm count diverges and the empirical second moments converge to G_a. Its asymptotic bias under the common reference law P_X is therefore $\|x^\top(\gamma_a - \gamma_{a_0})\|_{L^2(P_X)}$. If assignment is independent of context, every arm has the same context law, hence the same G_a and γ_a, and the projection terms cancel. Under action-dependent sampling they can differ, so a nonlinear f outside the fitted linear class can produce a nonzero contrast bias. This is a property of the specified separate-arm linear comparator, not of the reward objective: Eq. (3) shows that a correctly specified reward learner and the CATE learner share the same constrained optimum.

For part (c), write $m_t := r_t + \tau_{A_t}(X_t)$, so $s_t^{\hat{f}} = u_t(m_t + \xi_t)$ and, using $\mathbb{E}[\xi_t \mid X_t, A_t] = 0$,

$$\begin{aligned}\mathrm{Cov}(s_t^{\hat{f}} \mid X_t, \mathcal{H}_{t-1}) \\ = \mathbb{E}[u_t u_t^\top \sigma^2(X_t, A_t)] + \mathrm{Cov}(u_t m_t).\end{aligned}$$

The first term is the irreducible noise, independent of $\hat{f}$. For the second, $u_t m_t = r_t u_t + u_t\tau_{A_t}$ with r_t constant, so

$$\begin{aligned}\mathrm{Cov}(u_t m_t) = r_t^2 \underbrace{\mathrm{Cov}(u_t)}_{=M_t} \\ + 2r_t \underbrace{\mathrm{Cov}(u_t, u_t\tau_{A_t})}_{=\mathbb{E}[u_t u_t^\top \tau_{A_t}]} \\ + \mathrm{Cov}(u_t\tau_{A_t}),\end{aligned}$$

using $\mathbb{E}[u_t] = 0$, which is Eq. (25); only the first two terms depend on $\hat{f}$ through r_t. Taking traces and comparing the raw learner ($\hat{f} \equiv 0$, $r_t = f(X_t)$) with the oracle ($\hat{f} = f$, $r_t = 0$), the common noise and $\mathrm{Cov}(u_t\tau_{A_t})$ terms cancel, leaving

$$\begin{aligned}\mathrm{tr}\,\mathrm{Cov}(s_t^0) - \mathrm{tr}\,\mathrm{Cov}(s_t^f) \\ = f(X_t)^2\,\mathrm{tr}(M_t) + 2f(X_t)\,\mathbb{E}[\|u_t\|_2^2\,\tau_{A_t}],\end{aligned}$$

which is Eq. (26). This scalar is positive whenever $|f(X_t)|\,\mathrm{tr}(M_t) > 2\,|\mathbb{E}[\|u_t\|_2^2\tau_{A_t}]|$; since $|\tau_{A_t}| \le R_{\max}$ gives $|\mathbb{E}[\|u_t\|_2^2\tau_{A_t}]| \le R_{\max}\mathrm{tr}(M_t)$, the condition $|f(X_t)| > 2R_{\max}$ suffices when $\mathrm{tr}(M_t) > 0$. Thus residualizing the baseline strictly reduces the total score variance under these two conditions. If $\mathrm{tr}(M_t) = 0$, then $u_t = 0$ almost surely conditionally and both score covariances vanish. The matrix reduction is not one-directional in general because of the cross term $2r_t\mathbb{E}[u_t u_t^\top \tau_{A_t}]$; the total-variance statement is the correct one-directional guarantee. □

F Nonlinear Baselines and a Neural Variant

Assumption 5 restricts the CATEs to be linear in the raw context. The baseline f is already arbitrary: for any predictable $\hat{f}$, action-centering removes $f - \hat{f}$ from the conditional moment, so a nonlinear or misspecified baseline changes the confidence-radius constant but creates no cumulative bias. We next analyze a frozen-feature counterpart of the neural variant evaluated in Section 6.7. Motivated by neural-linear and deep-representation, shallow-exploration architectures [30, 31], it learns a neural representation from the independent offline sample, freezes that representation before online interaction, and updates only its linear head with action-centered feedback. Conditional on the offline sample, freezing fixes both the feature map and the head approximation used throughout the analysis. The resulting online features are predictable; predictability alone would also permit features updated from past observations.

F.1 Frozen neural representation and algorithm

Let $\widehat{W}$ be fitted only on the offline data and define an m-dimensional one-hidden-layer representation

$$\phi(x) := m^{-1/2} \tanh(\widehat{W}x + \widehat{b}) \in \mathbb{R}^m.$$

The normalization gives $\|\phi(x)\|_2 \le 1$; more generally, write $\|\phi(x)\|_2 \le L_\phi$. For the priced arms define

$$z_t^\phi(a) := e_a \otimes \phi(X_t) \in \mathbb{R}^{D_\phi}, \qquad D_\phi := Km, \qquad \bar{z}_t^\phi := \sum_a \pi_t(a) z_t^\phi(a).$$

The control block remains zero. We quantify the quality of the learned representation by a uniform approximation error for a bounded linear head:

$$\begin{aligned} &\tau_a(x) = z^\phi(x,a)^\top \theta_\phi^\star + \varepsilon_a^\phi(x), \qquad \|\theta_\phi^\star\|_2 \le S_\phi, \\ &\sup_{x \in \mathrm{supp}(P_X),\, a \in \mathcal{A}} |\varepsilon_a^\phi(x)| \le \varepsilon_\phi, \end{aligned} \tag{F.1}$$

where $\varepsilon_{a_0}^\phi \equiv 0$. Thus $\varepsilon_\phi = 0$ is exact realizability in a nonlinear, data-dependent feature space, not linearity in the raw context. The analysis treats ε_ϕ as a certified input and does not infer an approximation or finite-sample learning rate from network width or optimization.

For $\varepsilon_\phi = 0$, Eq. (F.2) is exactly the active-set rule of Algorithm 3 in the learned feature space. For $\varepsilon_\phi > 0$, the additive slack prevents elimination caused solely by representation error. As in the main paper, the farthest-pair rule is the analyzed selector. The experiment uses a Thompson-style distribution supported on the active set and calibrated confidence widths. The theorem below applies to such a selector only when it uses the stated active set and confidence radius and satisfies the coverage condition; the reported experiment is not presented as a theorem-certified run.

F.2 Confidence set and regret theorem

The offline sample is independent of the online interaction. We therefore state the result conditional on the fitted representation. This cleanly separates the online regret argument from any procedure-specific guarantee for learning ϕ.

Assumption F.1 (Frozen neural representation)**.** There is an event $\mathcal{E}_\phi$, measurable with respect to the independent offline sample, on which ϕ is fixed before round one, $0 < L_\phi < \infty$, $\|\phi(X_t)\|_2 \le L_\phi$ almost surely, Eq. (F.1) holds, and the projected warm start satisfies

$$\|\hat{\theta}_\phi^{(0)} - \theta_\phi^\star\|_2 \le \bar{\eta}_\phi.$$

Here $\mathbb{P}(\mathcal{E}_\phi) \ge 1 - \delta_{\mathrm{off}}$ and $0 \le \bar{\eta}_\phi \le 2S_\phi$. The deterministic choice $\bar{\eta}_\phi = 2S_\phi$ is always valid on the representation event because both $\theta_\phi^\star$ and $\hat{\theta}_\phi^{(0)}$ lie in the radius-S_ϕ ball. A sharper radius requires a representation-learner-specific offline analysis.

Assumption F.2 (Neural active-set selection)**.** Conditional on the frozen representation, the online selector draws a $\mathcal{G}_t$-measurable distribution π_t supported on $\mathcal{B}_t^\phi$ such that, for a constant $\kappa_0 > 0$ and every $a, a' \in \mathcal{B}_t^\phi$,

$$\|z_t^\phi(a) - z_t^\phi(a')\|^2_{(V_{t-1}^\phi)^{-1}} \le \kappa_0^{-1}\, \mathbb{E}\Big[\|u_t^\phi\|^2_{(V_{t-1}^\phi)^{-1}} \mid \mathcal{G}_t\Big]. \tag{F.3}$$

The farthest-pair rule in Algorithm F.1 satisfies this condition with $\kappa_0 = 1/4$ by Lemma A.2.

Set

$$\begin{aligned} q_T^\phi &= 4D_\phi \log(9T) + 8\log(20T/\delta), \\ h_T^\phi &= 27 D_\phi \log(1 + 2T/D_\phi) + 54 \log(20T/\delta), \end{aligned}$$

Algorithm F.1 Neural DR-CATE-CBwK with the analyzed farthest-pair rule.

Require: Logged data $\mathcal{D}$, budget B, horizon T, hidden width m, and a valid representation-error bound ε_ϕ.

Cross-fit the nuisances and form the DR pseudo-outcomes $\psi_i(a)$ by Eq. (6).

Fit the multi-output network $\widehat{\tau}_a(x) = v_a^\top \phi_{\widehat{W}}(x)$ to offline contrast targets, such as $\psi_i(a)$; freeze $\phi := \phi_{\widehat{W}}$, stack the fitted heads into $\widetilde{\theta}_\phi^{(0)}$, and fit the clipped baseline $\hat{f}$ as in Algorithm 2.

Set $\hat{\theta}_\phi^{(0)} = \Pi_{\{\|\theta\|_2 \le S_\phi\}}(\widetilde{\theta}_\phi^{(0)})$ and clip the offline dual initialization to $\lambda_1 \in [0, L_\phi S_\phi / c_{\min}]$.

Set $V_0^\phi = \alpha I_{D_\phi}$, $b_0^\phi = \alpha \hat{\theta}_\phi^{(0)}$, $\hat{\theta}_{\phi,0} = \hat{\theta}_\phi^{(0)}$, and $S_0 = 0$.

for $t = 1, \dots, T$ **do**

  Observe X_t and form the affordable set $\mathcal{A}_t^{\text{aff}}$ as in Algorithm 1.

  Set $\hat{s}_t^\phi(a) = z_t^\phi(a)^\top \hat{\theta}_{\phi,t-1} - \lambda_t c(X_t, a)$ and form

$$\mathcal{B}_t^\phi = \Big\{ a \in \mathcal{A}_t^{\text{aff}} : \hat{s}_t^\phi(a') - \hat{s}_t^\phi(a) \le 2\beta_t^\phi \|z_t^\phi(a') - z_t^\phi(a)\|_{(V_{t-1}^\phi)^{-1}} + 2\varepsilon_\phi, \quad \forall a' \in \mathcal{A}_t^{\text{aff}} \Big\}. \tag{F.2}$$

  If $\mathcal{B}_t^\phi$ is a singleton, play its action. Otherwise, put probability $1/2$ on a farthest pair in $\mathcal{B}_t^\phi$ under $\|\cdot\|_{(V_{t-1}^\phi)^{-1}}$ and draw $A_t \sim \pi_t$.

  Observe Y_t, update spend, and set $u_t^\phi = z_t^\phi(A_t) - \bar{z}_t^\phi$ and $g_t = Y_t - \hat{f}(X_t)$.

  Update

$$V_t^\phi = V_{t-1}^\phi + u_t^\phi (u_t^\phi)^\top, \qquad b_t^\phi = b_{t-1}^\phi + u_t^\phi g_t,$$
$$\hat{\theta}_{\phi,t} = \operatorname*{arg\,min}_{\|\theta\|_2 \le S_\phi} \|\theta - (V_t^\phi)^{-1} b_t^\phi\|_{V_t^\phi}.$$

  Set $\lambda_{t+1} = [\lambda_t + \eta_t(c(X_t, A_t) - \rho)]_+$.

end for

$\alpha = 4L_\phi^2 q_T^\phi$, and

$$\beta_t^\phi := (\Xi + 2F + L_\phi S_\phi)\sqrt{h_T^\phi} + \sqrt{\alpha}\,\bar{\eta}_\phi + \varepsilon_\phi \sqrt{t}. \tag{F.4}$$

Lemma F.1 (Neural action-centered confidence set)**.** *Under Assumptions 2, 6, and F.1, conditional on $\mathcal{E}_\phi$, with probability at least $1 - \delta/5$ simultaneously for $0 \le t \le T$,*

$$\|\hat{\theta}_{\phi,t} - \theta_\phi^\star\|_{V_t^\phi} \le \beta_t^\phi.$$

Theorem F.2 (Regret of the frozen neural variant)**.** *Suppose Assumptions 1, 2, 6, F.1, and F.2 hold. Run Algorithm F.1 with Eq.* (F.4) *and $\eta_t = \eta_0/\sqrt{T}$. Conditional on $\mathcal{E}_\phi$, with probability at least $1 - \delta$,*

$$R(T) \le \underbrace{C_1 D_\phi \sqrt{T} \log \tfrac{KT}{\delta}}_{R_{\text{online}}} + \underbrace{C_2 \sqrt{T}}_{R_{\text{dual}}} + \underbrace{C_3 D_\phi \sqrt{T}\,\bar{\eta}_\phi \log \tfrac{KT}{\delta}}_{R_{\text{warm}}} + \underbrace{C_4 \left(\varepsilon_\phi T \left[1 + \sqrt{D_\phi \log\left(1 + \frac{4L_\phi^2 T}{\alpha D_\phi} \right)} \right] + \frac{\varepsilon_\phi^2 T}{\sqrt{\alpha}} \right)}_{R_{\text{approx}}}. \tag{F.5}$$

The constants depend on Ξ, F, $c_{\max}$, $c_{\min}$, ρ, L_ϕ, S_ϕ, κ_0, η_0, and $\log(1/\delta)$, but not on T, m, K, or ε_ϕ. As in Theorem 1, the guarantee is for a fixed positive budget rate and is not uniform as $B/T \to 0$. Under exact neural realizability, $\varepsilon_\phi = 0$, it gives

$$R(T) = \widetilde{O}\Big(Km\sqrt{T}\Big) + \widetilde{O}\Big(Km\sqrt{T}\,\bar{\eta}_\phi\Big).$$

Thus the analyzed frozen-feature algorithm has the same horizon dependence as the linear-CATE algorithm, with raw dimension d replaced by hidden dimension m, holding L_ϕ and S_ϕ fixed. Unconditionally, the bound holds with probability at least $1 - \delta - \delta_{\text{off}}$.

The approximation term has two sources. The additive part is the direct discrepancy between the true CATE and its linear head, while the square-root factor arises when the misspecified head is estimated from action-centered feedback.

Two regimes should be distinguished. For fixed D_ϕ and bounded $\bar{\eta}_\phi$, the condition $\varepsilon_\phi = o((D_\phi \log T)^{-1/2})$ is sufficient only for sublinear regret, $R_{\text{approx}} = o(T)$; it does not preserve the statistical rate. To keep the approximation contribution no larger than the semiparametric term $\widetilde{O}(D_\phi\sqrt{T})$, and hence retain the overall $\widetilde{O}(D_\phi\sqrt{T})$ rate, one needs the stronger $\varepsilon_\phi = \widetilde{O}(\sqrt{D_\phi/T})$. If D_ϕ grows with T, one also needs $D_\phi(1+\bar{\eta}_\phi)\log(KT/\delta) = o(\sqrt{T})$. A fixed nonzero approximation error can produce linear regret without additional structure, consistent with known sensitivity to feature misspecification [32]. Assumption F.1 therefore keeps the representation guarantee explicit rather than attributing a finite-sample rate to the neural optimizer.

F.3 Proof of Lemma F.1

Proof. Condition throughout on $\mathcal{E}_\phi$, so ϕ and $\hat{\theta}_\phi^{(0)}$ satisfy the required offline bounds before online interaction. The claim at $t = 0$ follows from $V_0^\phi = \alpha I$ and the warm-start bound. For $t \geq 1$, write $\varepsilon_t^\phi = \varepsilon_{A_t}^\phi(X_t)$ and

$$\zeta_t^0 := f(X_t) - \hat{f}(X_t) + (\bar{z}_t^\phi)^\top \theta_\phi^\star + \xi_t.$$

By Eq. (F.1),

$$g_t = (u_t^\phi)^\top \theta_\phi^\star + \zeta_t^0 + \varepsilon_t^\phi.$$

Conditional on $\mathcal{G}_t = \sigma(\mathcal{H}_{t-1}, X_t)$, the non-noise part of ζ_t^0 is predictable and ξ_t is independent of the sampled action. Hence ζ_t^0 is conditionally independent of u_t^ϕ, $\|u_t^\phi\|_2 \leq 2L_\phi$, and $|\zeta_t^0| \leq \Xi + 2F + L_\phi S_\phi$. The same BOSE self-normalized inequality used in Lemma A.1 gives, simultaneously over t,

$$\left\| \sum_{s \leq t} u_s^\phi \zeta_s^0 \right\|_{(V_t^\phi)^{-1}} \leq (\Xi + 2F + L_\phi S_\phi)\sqrt{h_T^\phi}. \tag{F.6}$$

Also, $V_t^\phi \succeq \alpha I$ implies

$$\left\| \alpha(\hat{\theta}_\phi^{(0)} - \theta_\phi^\star) \right\|_{(V_t^\phi)^{-1}} \leq \sqrt{\alpha}\, \bar{\eta}_\phi.$$

It remains to control the action-dependent approximation error, for which BOSE conditional independence is unavailable. Let U_t have rows $(u_s^\phi)^\top$ and let $\varepsilon_{1:t}^\phi = (\varepsilon_1^\phi, \ldots, \varepsilon_t^\phi)^\top$. Since $V_t^\phi = \alpha I + U_t^\top U_t$,

$$\begin{aligned} \left\| \sum_{s \leq t} u_s^\phi \varepsilon_s^\phi \right\|_{(V_t^\phi)^{-1}}^2 &= (\varepsilon_{1:t}^\phi)^\top U_t (V_t^\phi)^{-1} U_t^\top \varepsilon_{1:t}^\phi \\ &\leq \|\varepsilon_{1:t}^\phi\|_2^2 \leq t\varepsilon_\phi^2. \end{aligned}$$

If σ_j are the singular values of U_t, the nonzero eigenvalues of $U_t(\alpha I + U_t^\top U_t)^{-1}U_t^\top$ are $\sigma_j^2/(\alpha + \sigma_j^2) \in [0, 1)$, which proves the matrix inequality.

For the unprojected least-squares estimate $\widetilde{\theta}_{\phi,t}$, the normal equations give

$$V_t^\phi(\widetilde{\theta}_{\phi,t} - \theta_\phi^\star) = \alpha(\hat{\theta}_\phi^{(0)} - \theta_\phi^\star) + \sum_{s \leq t} u_s^\phi \zeta_s^0 + \sum_{s \leq t} u_s^\phi \varepsilon_s^\phi.$$

The triangle inequality and the preceding three bounds yield $\|\widetilde{\theta}_{\phi,t} - \theta_\phi^\star\|_{V_t^\phi} \leq \beta_t^\phi$. Projection onto the radius-S_ϕ ball in the V_t^ϕ norm cannot increase the distance to $\theta_\phi^\star$, which belongs to that ball. This proves the lemma. □

F.4 Proof of Theorem F.2

Proof. All expectations and probabilities below are conditional on $\mathcal{E}_\phi$. We follow the proof of Theorem 1 and make the effects of representation error explicit. When $\varepsilon_\phi = 0$, the argument is, on $\mathcal{E}_\phi$, the linear-CATE proof with (X_t, L, S, D) replaced by $(\phi(X_t), L_\phi, S_\phi, D_\phi)$. The additional steps below control the action-dependent approximation residual, preserve the true optimum under elimination, and track the resulting increase in the dual queue. Set $G := c_{\max} + \rho$ and $R_{\max,\phi} := L_\phi S_\phi + \varepsilon_\phi$, so $|\tau_a(X_t)| \leq R_{\max,\phi}$.

If $\varepsilon_\phi > 1$, the deterministic bound $R(T) \leq 2R_{\max,\phi}T \leq 2(L_\phi S_\phi + 1)\varepsilon_\phi T$ proves Eq. (F.5) after increasing C_4. In the remainder, assume $0 \leq \varepsilon_\phi \leq 1$.

We first show that the true affordable optimum remains active. Define the true budget-adjusted score

$$s_t^\star(a) := \tau_a(X_t) - \lambda_t c(X_t, a) = z_t^\phi(a)^\top \theta_\phi^\star + \varepsilon_a^\phi(X_t) - \lambda_t c(X_t, a),$$

and let $a_t^\star$ maximize it over $\mathcal{A}_t^{\text{aff}}$. On the event of Lemma F.1, the bound at time $t-1$ and monotonicity of β_t^ϕ give, for every a, a',

$$\left|(z_t^\phi(a) - z_t^\phi(a'))^\top(\hat{\theta}_{\phi,t-1} - \theta_\phi^\star)\right| \le \beta_t^\phi \|z_t^\phi(a) - z_t^\phi(a')\|_{(V_{t-1}^\phi)^{-1}}.$$

For any a' the optimality of $a_t^\star$ and $|\varepsilon_{a'}^\phi(X_t) - \varepsilon_{a_t^\star}^\phi(X_t)| \le 2\varepsilon_\phi$ imply

$$\hat{s}_t^\phi(a') - \hat{s}_t^\phi(a_t^\star) \le \beta_t^\phi \|z_t^\phi(a') - z_t^\phi(a_t^\star)\|_{(V_{t-1}^\phi)^{-1}} + 2\varepsilon_\phi.$$

Therefore $a_t^\star \in \mathcal{B}_t^\phi$. Since the sampled action also lies in $\mathcal{B}_t^\phi$, Eq. (F.2) and one more use of the confidence inequality yield

$$s_t^\star(a_t^\star) - s_t^\star(A_t) \le 3\beta_t^\phi \|z_t^\phi(a_t^\star) - z_t^\phi(A_t)\|_{(V_{t-1}^\phi)^{-1}} + 4\varepsilon_\phi. \tag{F.7}$$

We next bound cumulative regret against the affordable optimum. By Assumption F.2,

$$\|z_t^\phi(a_t^\star) - z_t^\phi(A_t)\|^2_{(V_{t-1}^\phi)^{-1}} \le \kappa_0^{-1} w_t^\phi, \quad w_t^\phi := \mathbb{E}[\|u_t^\phi\|^2_{(V_{t-1}^\phi)^{-1}} \mid \mathcal{G}_t].$$

To control the predictable potentials, set $\Delta_t^\phi := \|u_t^\phi\|^2_{(V_{t-1}^\phi)^{-1}}$ and $b := 4L_\phi^2/\alpha \le 1$. Then $0 \le \Delta_t^\phi \le b$ and $\mathbb{E}[\Delta_t^\phi \mid \mathcal{G}_t] = w_t^\phi$. Applying the exponential-supermartingale argument of Eq. (21) to the frozen neural features gives, with probability at least $1 - \delta/5$,

$$\begin{aligned}\sum_{t=1}^T w_t^\phi &\le 2\sum_{t=1}^T \Delta_t^\phi + 2b\log(5/\delta) \\ &= O\left(D_\phi \log\left(1 + \frac{4L_\phi^2 T}{\alpha D_\phi}\right) + \log(1/\delta)\right).\end{aligned}$$

Here the realized sum is bounded by the elliptical potential inequality,

$$\sum_{t=1}^T \Delta_t^\phi \le 2D_\phi \log\left(1 + \frac{4L_\phi^2 T}{\alpha D_\phi}\right).$$

Because β_t^ϕ is nondecreasing, summing Eq. (F.7) and applying Cauchy-Schwarz gives

$$\begin{aligned}R_{\text{primal}}^{\text{aff}}(T) &\le 3\kappa_0^{-1/2}\beta_T^\phi \sqrt{T\sum_t w_t^\phi} + 4\varepsilon_\phi T \\ &= \widetilde{O}(D_\phi\sqrt{T}) + \widetilde{O}(D_\phi\sqrt{T}\,\bar{\eta}_\phi) + 4\varepsilon_\phi T \\ &\quad + O\left(\varepsilon_\phi T\sqrt{D_\phi \log\left(1 + \frac{4L_\phi^2 T}{\alpha D_\phi}\right)}\right).\end{aligned}$$

This is where the approximation term enters the bandit rate. No term involving $\|\hat{f} - f\|T$ appears: the shared baseline remains in ζ_t^0, where action-centering removes its conditional mean exactly.

We now control the dual variable, the prefix spend, and the affordable-set restriction, the three places where the representation error ε_ϕ enters the budget analysis. Throughout, recall $\alpha = 4L_\phi^2 q_T^\phi \ge 4L_\phi^2 > 0$, $G = c_{\max} + \rho$, and $R_{\max,\phi} = L_\phi S_\phi + \varepsilon_\phi$.

The dual term itself is unchanged. As in the derivation of Eq. (19), the update $\lambda_{t+1} = [\lambda_t + \eta(c(X_t, A_t) - \rho)]_+$ with $\eta = \eta_0/\sqrt{T}$ satisfies $\lambda_{t+1}^2 \le \lambda_t^2 + 2\eta\lambda_t(c(X_t, A_t) - \rho) + \eta^2 G^2$, so telescoping gives

$$R_{\text{dual}}(T) = -\sum_t \lambda_t(c(X_t, A_t) - \rho) \le \frac{\lambda_1^2}{2\eta} + \frac{\eta T G^2}{2} = O(\sqrt{T})$$

deterministically, independent of ε_ϕ and of the queue range.

The queue is uniformly bounded, with an ε_ϕ-dependent range. Projection onto the radius-S_ϕ ball and $V_{t-1}^\phi \succeq \alpha I$ give, for every affordable a, $|z_t^\phi(a)^\top \hat{\theta}_{\phi,t-1}| \le L_\phi S_\phi$ and $\|z_t^\phi(a)\|_{(V_{t-1}^\phi)^{-1}} \le L_\phi/\sqrt{\alpha}$. Since the free control has $z_t^\phi(a_0) = 0$, $c(X_t, a_0) = 0$, and $\varepsilon_{a_0}^\phi \equiv 0$, the rule Eq. (F.2) eliminates every priced action against a_0 once

$$\lambda_t > \bar{\lambda}_\phi := \frac{L_\phi S_\phi + 2\beta_T^\phi L_\phi/\sqrt{\alpha} + 2\varepsilon_\phi}{c_{\min}}.$$

On such a round $A_t = a_0$ and $\lambda_{t+1} = [\lambda_t - \eta\rho]_+ \le \lambda_t$, while any other round raises the queue by at most ηG. With the initialization $\lambda_1 \le L_\phi S_\phi / c_{\min}$, induction gives $0 \le \lambda_t \le \Lambda_\phi$ for all t, where

$$\Lambda_\phi := \max\{\lambda_1,\ \bar{\lambda}_\phi + \eta G\}.$$

Using $\bar{\eta}_\phi \le 2S_\phi$ and $\varepsilon_\phi \le 1$ in Eq. (F.4), $\beta_T^\phi/\sqrt{\alpha} = O(1) + O(\varepsilon_\phi\sqrt{T/\alpha})$, hence

$$\Lambda_\phi = O(1) + O(\varepsilon_\phi\sqrt{T/\alpha}).$$

The prefix spend obeys a deterministic bound. The nonnegative update gives $\lambda_{t+1}/\eta \ge \lambda_t/\eta + (c(X_t, A_t) - \rho)$, so telescoping and $0 \le \lambda_{t+1} \le \Lambda_\phi$ yield

$$S_t = \sum_{s \le t} c(X_s, A_s) \le t\rho + \frac{\lambda_{t+1} - \lambda_1}{\eta} \le t\rho + \frac{\Lambda_\phi}{\eta},$$

$$\frac{\Lambda_\phi}{\eta} = O(\sqrt{T}) + O\Big(\frac{\varepsilon_\phi T}{\sqrt{\alpha}}\Big).$$

Thus $B - S_{t-1} > c_{\max}$, i.e. $\mathcal{A}_t^{\text{aff}} = \mathcal{A}$, on all but the last

$$m_\phi := \Big\lceil \frac{\Lambda_\phi/\eta + c_{\max}}{\rho} \Big\rceil = O(\sqrt{T}) + O\Big(\frac{\varepsilon_\phi T}{\sqrt{\alpha}}\Big)$$

rounds.

The affordable optimum matches the fluid benchmark except on those rounds. The master decomposition Eq. (18) scores the global optimum over $\mathcal{A}$, whereas Eq. (F.7) bounds the gap against the affordable optimum $a_t^\star$ over $\mathcal{A}_t^{\text{aff}}$. On the $T - m_\phi$ rounds with $\mathcal{A}_t^{\text{aff}} = \mathcal{A}$ the two optima coincide; on each of the at most m_ϕ remaining rounds the free control is affordable with adjusted score zero, so replacing the global optimum by $a_t^\star$ costs at most $R_{\max,\phi}$. The affordable-set restriction therefore adds at most

$$R_{\max,\phi}\, m_\phi = O(\sqrt{T}) + O(\varepsilon_\phi\sqrt{T}) + O(\varepsilon_\phi T/\sqrt{\alpha}) + O(\varepsilon_\phi^2 T/\sqrt{\alpha}).$$

The master martingale difference in Eq. (18) is bounded by $2(R_{\max,\phi} + \Lambda_\phi G)$, so Azuma-Hoeffding contributes $O((R_{\max,\phi} + \Lambda_\phi G)\sqrt{T\log(1/\delta)}) = \widetilde{O}(\sqrt{T}) + O(\varepsilon_\phi T/\sqrt{\alpha})$. Collecting the preceding four displays, every ε_ϕ-free term is $\widetilde{O}(\sqrt{T})$ and joins $C_2\sqrt{T}$. The ε_ϕ-dependent contributions are $\varepsilon_\phi\sqrt{T}$, $\varepsilon_\phi T/\sqrt{\alpha}$, and $\varepsilon_\phi^2 T/\sqrt{\alpha}$; using $\sqrt{T} \le T$ and $1/\sqrt{\alpha} \le 1/(2L_\phi)$, the first two are $O(\varepsilon_\phi T)$, with constants allowed to depend on L_ϕ, and together with the $O(\varepsilon_\phi T\sqrt{D_\phi \log(1 + 4L_\phi^2 T/(\alpha D_\phi))})$ term from $R_{\text{primal}}^{\text{aff}}$ and the residual $\varepsilon_\phi^2 T/\sqrt{\alpha}$ they form R_{approx} in Eq. (F.5). The initial case distinction covers all $\varepsilon_\phi \ge 0$. The free control is affordable on every round, so $\sum_t c(X_t, A_t) \le B$ holds pathwise on every sample path.

Inserting these bounds into the weak-duality master decomposition Eq. (18) completes the argument. Conditional on $\mathcal{E}_\phi$, the confidence event, master martingale event, and predictable-to-realized potential event each fail with probability at most $\delta/5$; the dual and pathwise-feasibility bounds are deterministic. A union bound proves Eq. (F.5) with probability at least $1 - \delta$. Finally, $\mathbb{P}(\mathcal{E}_\phi^c) \le \delta_{\text{off}}$ gives the unconditional probability. Setting $\varepsilon_\phi = 0$ removes every approximation term and proves the exact-realizability statement. □

The theorem applies to any predictable representation fixed before online interaction. Assumption F.1 supplies ε_ϕ and $\bar{\eta}_\phi$ as certified offline inputs; the result does not derive them from a particular network width, sample size, or optimizer. Unlike NeuralUCB and Neural Thompson Sampling [33, 34], it also does not update the representation during interaction.

The next section analyzes a complementary online construction: the nonlinear network is retrained after each round, while initialization tangent features provide its confidence geometry.

G An Online Action-Centered Neural-Tangent Extension

This section develops a separate neural-tangent extension in which the network is retrained after every observation, while exploration uses gradients frozen at initialization. The construction differs from the frozen representation of Appendix F: the complete nonlinear network supplies the online score, while its confidence geometry is measured by the information gain of an action-centered neural tangent kernel (NTK). The key device is balanced farthest-pair sampling. It converts action-centering into a Rademacher pseudo-outcome with a predictable tangent contrast, which permits a kernel self-normalized argument without applying an ordinary linear-bandit theorem to a feature observed only after action sampling.

The result gives an online neural-network algorithm with a conditional NTK regret guarantee for nonlinear CATE functions in the initialization tangent-feature class. It combines causal action-centering, nonlinear network scoring, fixed centered-kernel information gain, and a hard budget constraint. The regret theorem isolates one direct centered-loss coupling event between nonlinear pair predictions and the fixed-NTK ridge predictor. This avoids requiring a network to track a ridge estimator assembled from different historical parameter iterates. It is not an end-to-end finite-width theorem until that event is verified from explicit architecture, width, learning-rate, and optimization assumptions.

G.1 Inherited setting and NTK-specific scope

We retain the action set, outcome decomposition, cost conditions, budget rate, policy value $V(\pi)$, policy cost $C(\pi)$, fluid benchmark $\mathrm{OPT}_{\mathrm{LP}}(\rho)$, and regret $R(T)$ from Section 3. Thus Assumptions 1, 2, and 6 continue to hold. The prefit baseline $\hat{f}$ is constructed as in Section 4, independently of the online sample and clipped to $[-F, F]$. Only linear CATE realizability is replaced by the nonlinear condition below.

Let $\mathcal{H}_0^{\mathrm{N}}$ contain the initialization θ_0, the prefit baseline $\hat{f}$, and any other pre-online algorithmic randomness, all independent of the online sequence, and define

$$\mathcal{H}_t^{\mathrm{N}} := \sigma(\mathcal{H}_0^{\mathrm{N}}, X_1, A_1, Y_1, \ldots, X_t, A_t, Y_t).$$

The conditional statements in Assumption 6 are understood with this augmented history. At the end of round t, the algorithmic state $(\theta_t, V_t^0, b_t^0, \widehat{w}_t^0, \lambda_{t+1}, S_t)$ is $\mathcal{H}_t^{\mathrm{N}}$-measurable.

Assumption G.1 (Bounded nonlinear CATE)**.** The treatment contrasts satisfy

$$\sup_{x \in \mathrm{supp}(P_X), a \in \mathcal{A}} |\tau_a(x)| \le R_{\mathrm{NTK}}.$$

The extension differs from the frozen neural variant because the complete network is retrained online, although confidence widths retain the initialization tangent geometry. It also differs from Neural Thompson Sampling. Neural Thompson Sampling regresses observed rewards on single-action network outputs and samples Gaussian reward scores. The method below regresses action-centered pairwise CATE observations on tangent contrasts and uses a balanced farthest-pair rule. The symmetry of that rule, rather than posterior sampling, removes the shared baseline exactly.

G.2 Online network and balanced action-centering

Let $h : \mathcal{A} \times \mathcal{X} \times \mathbb{R}^p \to \mathbb{R}$ be an action-conditioned L_{net}-layer fully connected network of width m with shared parameters $\theta \in \mathbb{R}^p$. The action can be encoded by a one-hot input or an equivalent multi-output head. We hard-code $h(a_0, x; \theta) = 0$. The initialization θ_0 is an algorithm input independent of the data and satisfies $h(a, x; \theta_0) = 0$. For example, this identity follows by duplicating each initialized hidden feature with opposite output weights. The theorem requires no further initialization law beyond Assumption G.2. Define the initialization tangent feature

$$g_0(a, x) := m^{-1/2} \nabla_\theta h(a, x; \theta_0), \qquad g_0(a_0, x) := 0. \tag{G.1}$$

At the start of round t, the current parameters θ_{t-1} are $\mathcal{H}_{t-1}^{\mathrm{N}}$-measurable, but the confidence feature is $g_0(a) := g_0(a, X_t)$. Thus $g_0(a)$ is predictable after observing X_t and before drawing A_t. The network value $h(a, X_t; \theta_{t-1})$, rather than a frozen linear head, is the online CATE estimate.

Suppose a predictable active set $\mathcal{B}_t$ contains at least two actions. Choose a pair (a_t^+, a_t^-) from this set, draw an independent Rademacher sign $\sigma_t \in \{-1, +1\}$, and play a_t^+ when $\sigma_t = +1$ and a_t^- when $\sigma_t = -1$. Define

$$\begin{aligned} d_t^0 &:= g_0(a_t^+) - g_0(a_t^-), \\ \widetilde{Y}_t &:= 2\sigma_t \{Y_t - \hat{f}(X_t)\}. \end{aligned} \tag{G.2}$$

If the active set is the singleton $\{a\}$, set $a_t^+ = a_t^- = a$, play a, and set $d_t^0 = 0$ and $\widetilde{Y}_t = 0$. With $V_0^0 = \alpha_{\mathrm{NTK}} I$ and $b_0^0 = 0$, maintain the initialization-tangent ridge quantities

$$V_t^0 = V_{t-1}^0 + d_t^0 (d_t^0)^\top, \qquad b_t^0 = b_{t-1}^0 + d_t^0 \widetilde{Y}_t, \qquad \widehat{w}_t^0 = (V_t^0)^{-1} b_t^0. \tag{G.3}$$

These quantities are used in the proof and in the confidence radius. The action score itself remains the nonlinear network output.

After round t, retrain the full network from θ_0 for J gradient steps on the regularized pairwise loss

$$\mathcal{L}_t(\theta) := \frac{1}{2} \sum_{s \le t} \Big(h(a_s^+, X_s; \theta) - h(a_s^-, X_s; \theta) - \widetilde{Y}_s \Big)^2 + \frac{m \alpha_{\mathrm{NTK}}}{2} \|\theta - \theta_0\|_2^2. \tag{G.4}$$

The scaling connecting this objective to Eq. (G.3) is

$$w(\theta) := \sqrt{m}\,(\theta - \theta_0). \tag{G.5}$$

Because the initialization output is zero, first-order expansion of the pair output gives

$$h(a_s^+, X_s; \theta) - h(a_s^-, X_s; \theta) = (d_s^0)^\top w(\theta) + \mathrm{Rem}_s(\theta).$$

Moreover,

$$\frac{m \alpha_{\mathrm{NTK}}}{2} \|\theta - \theta_0\|_2^2 = \frac{\alpha_{\mathrm{NTK}}}{2} \|w(\theta)\|_2^2.$$

Discarding the Taylor remainders therefore gives the linearized objective

$$\mathcal{L}_t^0(w) := \frac{1}{2} \sum_{s \le t} \Big((d_s^0)^\top w - \widetilde{Y}_s \Big)^2 + \frac{\alpha_{\mathrm{NTK}}}{2} \|w\|_2^2, \qquad \arg\min_w \mathcal{L}_t^0(w) = \widehat{w}_t^0. \tag{G.6}$$

The last identity follows from the normal equations and shows that the ridge parameter is the exact optimizer of the initialization-linearized neural loss, not an unrelated auxiliary estimator. The next round therefore uses a newly trained nonlinear score, while retaining the initialization features for uncertainty. The guarantee remains an NTK guarantee because its realizability and confidence conditions are stated in tangent-feature space. Restarting from θ_0 and fitting all observations at every round is a theoretical construction chosen to simplify comparison with the initialization NTK. It uses $J \sum_{t=1}^{T} t = JT(T+1)/2$ example-gradient evaluations, apart from network-evaluation costs. A warm-started implementation would be cheaper, but requires a different stability lemma and is not analyzed here.

Given a symmetric pairwise confidence radius $r_t(a, a')$, later instantiated in Eq. (G.13), define the residual spend and affordable set by $S_{t-1} = \sum_{s<t} c(X_s, A_s)$ and $\mathcal{A}_t^{\mathrm{aff}} = \{a : c(X_t, a) \le B - S_{t-1}\}$. The budget-adjusted network score and active set are defined below. If the active set is empty, the algorithm plays the control. When the active set has at least two actions, its farthest distinct pair is also defined below. All maximization ties are resolved by a fixed deterministic ordering. Hence the active set and selected pair are measurable with respect to $\sigma(\mathcal{H}_{t-1}^{\mathrm{N}}, X_t)$.

$$\begin{aligned} \widehat{s}_t(a) &:= h(a, X_t; \theta_{t-1}) - \lambda_t c(X_t, a), \\ \mathcal{B}_t^{\mathrm{NTK}} &:= \Big\{ a \in \mathcal{A}_t^{\mathrm{aff}} : \widehat{s}_t(a') - \widehat{s}_t(a) \le 2 r_t(a, a'), \forall a' \in \mathcal{A}_t^{\mathrm{aff}} \Big\}, \\ (a_t^+, a_t^-) &\in \operatorname*{arg\,max}_{\substack{a, a' \in \mathcal{B}_t^{\mathrm{NTK}} \\ a \ne a'}} \|g_0(a) - g_0(a')\|_{(V_{t-1}^0)^{-1}}. \end{aligned} \tag{G.7}$$

The algorithm maintains two coupled objects. The network output supplies the nonlinear action score, while $(V_t^0, b_t^0, \widehat{w}_t^0)$ supplies fixed initialization confidence geometry used to certify network contrasts. The next propositions establish that balanced pairwise feedback targets the CATE difference and preserves the positive-semidefinite NTK structure.

Proposition G.1 (Exact action-centered pseudo-outcome)**.** *Let $\mathcal{G}_t = \sigma(\mathcal{H}_{t-1}^{\mathrm{N}}, X_t, a_t^+, a_t^-)$, and suppose Assumptions 2, 6, and G.1 hold. Then*

$$\mathbb{E}[\widetilde{Y}_t \mid \mathcal{G}_t] = \tau_{a_t^+}(X_t) - \tau_{a_t^-}(X_t). \tag{G.8}$$

Moreover,

$$\widetilde{Y}_t = \tau_{a_t^+}(X_t) - \tau_{a_t^-}(X_t) + \nu_t, \qquad \mathbb{E}[\nu_t \mid \mathcal{G}_t] = 0, \tag{G.9}$$

where $|\nu_t| \le R_{\mathrm{ac}} := 2(2F + R_{\mathrm{NTK}} + \Xi)$ almost surely.

Algorithm G.1 Action-Centered NTK-CBwK with balanced pairwise exploration.

Require: Budget B, horizon T, baseline estimate $\hat{f}$, initialized network $h(\cdot,\cdot;\theta_0)$, ridge α_{NTK}, and dual step η_0.
Require: Neural step size γ, gradient steps J, initial dual $\lambda_1 \geq 0$, and radius r_t from Eq. (G.13).
Set $\rho \leftarrow B/T$, $V_0^0 \leftarrow \alpha_{\mathrm{NTK}} I_p$, $b_0^0 \leftarrow 0$, $\widehat{w}_0^0 \leftarrow 0$, and $S_0 \leftarrow 0$.
for $t = 1, \ldots, T$ **do**
  Observe X_t and compute $g_0(a) = g_0(a, X_t)$ from Eq. (G.1) for every $a \in \mathcal{A}$.
  Set $\mathcal{A}_t^{\mathrm{aff}} \leftarrow \{a \in \mathcal{A} : c(X_t, a) \leq B - S_{t-1}\}$.
  Compute $\widehat{s}_t$ and $\mathcal{B}_t^{\mathrm{NTK}}$ using Eq. (G.7).
  if $\mathcal{B}_t^{\mathrm{NTK}} = \varnothing$ **then**
    Set $a_t^+ = a_t^- = A_t \leftarrow a_0$, $d_t^0 \leftarrow 0$, and $\widetilde{Y}_t \leftarrow 0$.
  else if $\mathcal{B}_t^{\mathrm{NTK}} = \{a\}$ **then**
    Set $a_t^+ = a_t^- = A_t \leftarrow a$, $d_t^0 \leftarrow 0$, and $\widetilde{Y}_t \leftarrow 0$.
  else
    Choose (a_t^+, a_t^-) by the farthest-pair rule in Eq. (G.7).
    Draw σ_t uniformly from $\{-1, +1\}$ and set $A_t \leftarrow a_t^+$ if $\sigma_t = +1$, and $A_t \leftarrow a_t^-$ otherwise.
  end if
  Observe Y_t and set $S_t \leftarrow S_{t-1} + c(X_t, A_t)$.
  if $a_t^+ \neq a_t^-$ **then**
    Set $d_t^0 \leftarrow g_0(a_t^+) - g_0(a_t^-)$ and $\widetilde{Y}_t \leftarrow 2\sigma_t\{Y_t - \hat{f}(X_t)\}$.
  end if
  Update $(V_t^0, b_t^0, \widehat{w}_t^0)$ by Eq. (G.3).
  Set $\theta_t^{(0)} \leftarrow \theta_0$.
  for $j = 0, \ldots, J-1$ **do**
    $\theta_t^{(j+1)} \leftarrow \theta_t^{(j)} - \gamma \nabla_\theta \mathcal{L}_t(\theta_t^{(j)})$.
  end for
  Set $\theta_t \leftarrow \theta_t^{(J)}$.
  Set $\lambda_{t+1} \leftarrow [\lambda_t + (\eta_0/\sqrt{T})\{c(X_t, A_t) - \rho\}]_+$.
end for

Proof. The singleton case has $a_t^+ = a_t^-$ and both sides of Eq. (G.8) are zero. Otherwise, write $r_t^f = f(X_t) - \hat{f}(X_t)$ and $\tau_t^{\pm} = \tau_{a_t^{\pm}}(X_t)$. Balanced sampling gives

$$\tau_{A_t}(X_t) = \frac{\tau_t^+ + \tau_t^-}{2} + \frac{\sigma_t}{2}(\tau_t^+ - \tau_t^-).$$

Substitution into $2\sigma_t(Y_t - \hat{f}(X_t))$ yields

$$\widetilde{Y}_t = \tau_t^+ - \tau_t^- + 2\sigma_t \left(r_t^f + \frac{\tau_t^+ + \tau_t^-}{2} + \xi_t \right).$$

Given $\mathcal{G}_t$ and a distinct pair, A_t is a one-to-one function of σ_t. Conditional independence of ξ_t from the sampled action therefore gives $\mathbb{E}[\sigma_t \xi_t \mid \mathcal{G}_t] = 0$. The remaining terms in parentheses are $\mathcal{G}_t$-measurable and $\mathbb{E}[\sigma_t \mid \mathcal{G}_t] = 0$. Hence the last term is a martingale difference. Clipping gives $|r_t^f| \leq 2F$, while $|\tau_t^{\pm}| \leq R_{\mathrm{NTK}}$ and $|\xi_t| \leq \Xi$, which proves the bound. □

Proposition G.1 is the main compatibility result between action-centering and NTK exploration. The random sign moves into the response, while the initialization contrast d_t^0 is fixed before sampling. Consequently, standard predictable-covariate kernel concentration applies to the sequence $(d_t^0, \widetilde{Y}_t)$. A general nonbalanced action law instead produces a post-action centered feature and would require a kernel-valued extension of the BOSE self-normalized inequality.

Proposition G.2 (Centering preserves the NTK structure)**.** *Let $K_0((x,a),(x',a')) = \langle g_0(a,x), g_0(a',x') \rangle$ be the initialization NTK. For pair contrasts $q = (x, a^+, a^-)$ define*

$$K_0^{\mathrm{ac}}(q, q') := \left\langle g_0(a^+, x) - g_0(a^-, x), g_0(a'^+, x') - g_0(a'^-, x') \right\rangle. \tag{G.10}$$

Then K_0^{ac} is positive semidefinite. For any finite collection of pairs, its Gram matrix is $CH_0C^\top$, where H_0 is the uncentered NTK Gram matrix over the participating action-context inputs and each row of C is the corresponding signed contrast.

Proof. For any coefficients $v_1, \dots, v_n$,

$$\sum_{i,j} v_i v_j K_0^{\mathrm{ac}}(q_i, q_j) = \left\| \sum_i v_i \{ g_0(a_i^+, x_i) - g_0(a_i^-, x_i) \} \right\|_2^2 \geq 0.$$

The matrix identity follows by expanding the two signed differences. Thus centering changes the kernel by a linear contrast operator and does not destroy positive semidefiniteness. □

G.3 NTK assumptions and confidence set

The following assumption states initialization-feature realizability and isolates the architecture-specific step that transfers fixed-NTK ridge confidence to the trained nonlinear pair predictor. Under the usual limiting-kernel interpretation, the first condition is the corresponding centered NTK RKHS condition. Unlike a moving-gradient formulation, no comparison among historical tangent geometries is needed.

For a deterministic radius R_θ, define the NTK neighborhood

$$\mathcal{N}_\theta := \{\theta : \|w(\theta)\|_2 \leq R_\theta\}.$$

For each $t \geq 1$, let $\bar{\theta}_t$ be a selected element of $\arg\min_{\theta \in \mathcal{N}_\theta} \mathcal{L}_t(\theta)$, and set $\bar{\theta}_0 = \theta_0$. Write $w_t = w(\theta_t)$ and $\bar{w}_t = w(\bar{\theta}_t)$ using Eq. (G.5). This reference definition does not assume convergence of the gradient-descent parameter sequence.

Assumption G.2 (Modular fixed-NTK verification conditions)**.** The bounds and error tolerances below are deterministic and nonnegative, with $L_g > 0$ and $\alpha_{\mathrm{NTK}} > 0$. For P_{θ_0}-almost every initialization, there is a vector $w^\star$, measurable with respect to θ_0 and therefore selected before online interaction, such that $\|w^\star\|_2 \leq S_{\mathrm{NTK}}$ and, for a deterministic $\epsilon_{\mathrm{init}}(m)$,

$$\left|\tau_a(x) - g_0(a,x)^\top w^\star\right| \leq \epsilon_{\mathrm{init}}(m)$$

for P_X-almost every x and every $a \in \mathcal{A}$.

Under the joint law of the initialization and online trajectory, there are events $\mathcal{E}_{\mathrm{env}}$, $\mathcal{E}_{\mathrm{lin}}$, $\mathcal{E}_{\mathrm{id}}$, and $\mathcal{E}_{\mathrm{opt}}$ whose failure probabilities are at most δ_{env}, δ_{lin}, δ_{id}, and δ_{opt}, respectively, with the following properties.

1. On $\mathcal{E}_{\mathrm{env}}$, the selected minimizers $\bar{\theta}_t$ exist, every gradient-descent iterate remains in $\mathcal{N}_\theta$, and initialization gradients and trained outputs satisfy
$$\sup_{a,x} \|g_0(a,x)\|_2 \leq L_g,$$
$$\sup_{t,a,x} |h(a,x;\theta_{t-1})| \leq R_h,$$
where the suprema use $t \leq T$, $a \in \mathcal{A}$, and $x \in \mathrm{supp}(P_X)$.
2. On $\mathcal{E}_{\mathrm{lin}}$, uniformly over $t \leq T$, $a, a' \in \mathcal{A}$, and $x \in \mathrm{supp}(P_X)$,
$$\left|h(a,x;\theta_{t-1}) - h(a',x;\theta_{t-1}) - \{g_0(a,x) - g_0(a',x)\}^\top w_{t-1}\right| \leq \epsilon_{\mathrm{lin}}(m,T).$$
3. On $\mathcal{E}_{\mathrm{opt}}$, uniformly over the same indices,
$$\left|\{g_0(a,x) - g_0(a',x)\}^\top (w_{t-1} - \bar{w}_{t-1})\right| \leq \epsilon_{\mathrm{opt}}(m,T,J).$$
4. On $\mathcal{E}_{\mathrm{id}}$, uniformly over the same indices,
$$\left|\{g_0(a,x) - g_0(a',x)\}^\top (\bar{w}_{t-1} - \widehat{w}_{t-1}^0)\right| \leq \epsilon_{\mathrm{id}}(m,T).$$
5. There is a deterministic capacity $\overline{\Gamma}_T^0$ such that, for every initialization in a P_{θ_0}-probability-one set and uniformly over all context and pair sequences of length at most T,
$$\log \frac{\det V_t^0}{\det(\alpha_{\mathrm{NTK}} I)} \leq \overline{\Gamma}_T^0.$$

Define

$$\begin{aligned} \mathcal{E}_{\mathrm{cpl}} &:= \mathcal{E}_{\mathrm{env}} \cap \mathcal{E}_{\mathrm{lin}} \cap \mathcal{E}_{\mathrm{id}} \cap \mathcal{E}_{\mathrm{opt}}, \\ \delta_{\mathrm{cpl}} &:= \delta_{\mathrm{env}} + \delta_{\mathrm{lin}} + \delta_{\mathrm{id}} + \delta_{\mathrm{opt}}, \\ \epsilon_{\mathrm{fw}}(m,T) &:= \epsilon_{\mathrm{lin}}(m,T) + \epsilon_{\mathrm{id}}(m,T), \\ \epsilon_{\mathrm{cpl}}(m,T,J) &:= \epsilon_{\mathrm{fw}}(m,T) + \epsilon_{\mathrm{opt}}(m,T,J). \end{aligned} \tag{G.11}$$

Then $\mathbb{P}(\mathcal{E}_{\mathrm{cpl}}) \geq 1 - \delta_{\mathrm{cpl}}$ by the union bound. Here ϵ_{lin} is the nonlinear Taylor remainder, ϵ_{id} is the argmin-stability or ridge-identification error relative to the exact optimizer of Eq. (G.6), and ϵ_{opt} is the residual from stopping after J steps.

The representer $w^\star$ and $\epsilon_{\text{init}}(m)$ are selected before online interaction and do not depend on online observations. The coupling event contains no moving-gradient ridge equivalence or dynamic Gram comparison. Its component bounds and the triangle inequality imply the direct nonlinear-to-ridge bound

$$\begin{aligned}&\left|h(a,x;\theta_{t-1}) - h(a',x;\theta_{t-1}) - \{g_0(a,x) - g_0(a',x)\}^\top \widehat{w}_{t-1}^0\right| \\ &\quad \le \epsilon_{\text{cpl}}(m,T,J).\end{aligned}$$

This is the finite-width coupling target required to transfer the fixed-NTK confidence bound to the nonlinear predictor.

For almost every initialization, approximation holds for P_X-almost every context. Independence of the initialization and the i.i.d. context sequence then implies that it holds at all finitely many realized contexts almost surely. Under exact initialization realizability, $\epsilon_{\text{init}}(m) = 0$.

For ReLU networks, the nonsmooth activation requires the sign-stability analysis used by NeuralUCB and Neural Thompson Sampling. For a smooth activation, the same role is played by uniform Hessian control in the neighborhood visited by gradient descent. Pairwise neural bandit analysis provides the closest template because its loss and confidence objects are also action contrasts. In either case, the width and optimization length must control ϵ_{fw} and ϵ_{opt} uniformly over the adaptive pair sequence.

Define $\Gamma_t^0 := \log\{\det(V_t^0)/\det(\alpha_{\text{NTK}} I)\}$, including $\Gamma_0^0 = 0$. For $t \ge 0$, let

$$\begin{aligned}\beta_t^{\text{NTK}} &:= R_{\text{ac}}\sqrt{\Gamma_t^0 + 2\log(1/\delta)} + \sqrt{\alpha_{\text{NTK}}} S_{\text{NTK}} + 2\epsilon_{\text{init}}(m)\sqrt{t}, \\ \epsilon_{m,J} &:= 2\epsilon_{\text{init}}(m) + \epsilon_{\text{cpl}}(m,T,J).\end{aligned} \tag{G.12}$$

In particular, $\beta_0^{\text{NTK}} = R_{\text{ac}}\sqrt{2\log(1/\delta)} + \sqrt{\alpha_{\text{NTK}}} S_{\text{NTK}}$. The $2\epsilon_{\text{init}}(m)\sqrt{t}$ term in β_t^{NTK} accumulates approximation residuals from the t historical regression targets. The separate $2\epsilon_{\text{init}}(m)$ in $\epsilon_{m,J}$ controls the two endpoints of the current candidate pair. For a candidate pair a, a', set

$$r_t(a,a') := \beta_{t-1}^{\text{NTK}} \|g_0(a) - g_0(a')\|_{(V_{t-1}^0)^{-1}} + \epsilon_{m,J}. \tag{G.13}$$

Lemma G.3 (Action-centered NTK contrast confidence)**.** *Under Assumptions 1, 2, 6, G.1, and G.2, there is an event $\mathcal{C}_\delta$ with probability at least $1-\delta$ such that, on $\mathcal{E}_{\text{cpl}} \cap \mathcal{C}_\delta$, simultaneously for every $t \le T$ and $a, a' \in \mathcal{A}$,*

$$\begin{aligned}&|\{\tau_a(X_t) - \tau_{a'}(X_t)\} - \{h(a,X_t;\theta_{t-1}) - h(a',X_t;\theta_{t-1})\}| \\ &\quad \le r_t(a,a').\end{aligned} \tag{G.14}$$

Proof. Under the original filtration and probability measure, Proposition G.1 gives the martingale difference ν_s. The predictable-covariate self-normalized inequality of Abbasi-Yadkori et al. [29] therefore defines an event $\mathcal{C}_\delta$ with $\mathbb{P}(\mathcal{C}_\delta) \ge 1-\delta$ on which, simultaneously over t,

$$\left\|\sum_{s\le t} d_s^0 \nu_s\right\|_{(V_t^0)^{-1}} \le R_{\text{ac}}\sqrt{\Gamma_t^0 + 2\log(1/\delta)}.$$

This concentration step is not conditioned on the possibly path-dependent event $\mathcal{E}_{\text{cpl}}$. On the intersection $\mathcal{C}_\delta \cap \mathcal{E}_{\text{cpl}}$, which has probability at least $1-\delta-\delta_{\text{cpl}}$, initialization approximation gives

$$\widetilde{Y}_s = (d_s^0)^\top w^\star + e_s + \nu_s, \qquad |e_s| \le 2\epsilon_{\text{init}}(m),$$

where d_s^0 is $\mathcal{G}_s$-measurable. The deterministic approximation residual satisfies

$$\left\|\sum_{s\le t} d_s^0 e_s\right\|_{(V_t^0)^{-1}} \le \|e_{1:t}\|_2 \le 2\epsilon_{\text{init}}(m)\sqrt{t}.$$

The first inequality follows because the nonzero eigenvalues of $D_t(\alpha_{\text{NTK}} I + D_t^\top D_t)^{-1} D_t^\top$ lie in $[0,1)$, where D_t has rows $d_s^{0\top}$. Finally, $\|\alpha_{\text{NTK}} w^\star\|_{(V_t^0)^{-1}} \le \sqrt{\alpha_{\text{NTK}}} S_{\text{NTK}}$. The normal equations in Eq. (G.3) therefore imply

$$\|\widehat{w}_t^0 - w^\star\|_{V_t^0} \le \beta_t^{\text{NTK}}.$$

For any candidate pair, add the two initialization-approximation residuals and apply Cauchy-Schwarz to $\{g_0(a) - g_0(a')\}^\top(\widehat{w}_{t-1}^0 - w^\star)$. The direct nonlinear coupling in Assumption G.2 adds $\epsilon_{\text{cpl}}(m,T,J)$, yielding Eq. (G.14). □

G.4 Regret theorem

Define the centered NTK effective dimension by

$$\widetilde{d}_{\mathrm{ac}} := \max\left\{1,\ \frac{\overline{\Gamma}_T^0}{\log(1+4L_g^2T/\alpha_{\mathrm{NTK}})}\right\}. \tag{G.15}$$

The deterministic capacity $\overline{\Gamma}_T^0$ takes a worst-case supremum over initialization-feature context and pair sequences. This choice avoids an additional probabilistic complexity event for the adaptive sequence and gives a deterministic envelope for the queue and martingale bounds. This is the pairwise analogue of the effective dimension in NeuralUCB and Neural Thompson Sampling. It measures the capacity of tangent contrasts over all candidate pairs; the deterministic supremum is not restricted to the realized active sets. It can be smaller than the effective dimension of the full un-centered action-context kernel. The bound depends on this information gain rather than directly on the raw parameter dimension. Under the direct coupling event, width enters only through the initialization kernel and the finite-width error.

Theorem G.4 (Conditional regret for the online neural-tangent variant)**.** *Suppose Assumptions 1, 2, 6, G.1, and G.2 hold. Run Algorithm G.1 with $\alpha_{\mathrm{NTK}} \geq 4L_g^2$ and $\lambda_1 \leq R_h/c_{\min}$. Write $\eta := \eta_0/\sqrt{T}$ for the constant dual step used by the algorithm. Let*

$$\begin{aligned}
\overline{\beta}_T &:= R_{\mathrm{ac}}\sqrt{\overline{\Gamma}_T^0 + 2\log(1/\delta)} + \sqrt{\alpha_{\mathrm{NTK}}}S_{\mathrm{NTK}} + 2\epsilon_{\mathrm{init}}(m)\sqrt{T},\\
\overline{r}_T &:= \frac{L_g\overline{\beta}_T}{\sqrt{\alpha_{\mathrm{NTK}}}} + \epsilon_{m,J},\\
\overline{\lambda}_T &:= \frac{R_h + 2\overline{r}_T}{c_{\min}},\\
\Lambda_T &:= \max\{\lambda_1, \overline{\lambda}_T + \eta(c_{\max}+\rho)\},\\
N_T^{\mathrm{aff}} &:= \left\lceil \frac{\Lambda_T/\eta + c_{\max}}{\rho} \right\rceil.
\end{aligned} \tag{G.16}$$

Let $\mathcal{C}_\delta$ be the concentration event from Lemma G.3, and let $\mathcal{Z}_\delta$ be the stopped Azuma-Hoeffding event defined in the proof below. With probability at least $1-2\delta-\delta_{\mathrm{cpl}}$, the event $\mathcal{E}_{\mathrm{cpl}} \cap \mathcal{C}_\delta \cap \mathcal{Z}_\delta$ holds and the regret against the fluid LP benchmark satisfies

$$\begin{aligned}
R(T) \leq\ & 3\beta_T^{\mathrm{NTK}}\sqrt{2T\Gamma_T^0} + 3\epsilon_{m,J}T\\
&+ \frac{\lambda_1^2}{2\eta} + \frac{\eta T(c_{\max}+\rho)^2}{2}\\
&+ C\{R_{\mathrm{NTK}} + \Lambda_T(c_{\max}+\rho)\}\\
&\quad\times\sqrt{T\log(1/\delta)} + R_{\mathrm{NTK}}N_T^{\mathrm{aff}},
\end{aligned} \tag{G.17}$$

for a numerical constant C. The affordable-action rule enforces $\sum_{t=1}^T c(X_t,A_t) \leq B$ on every sample path.

Proof. Let $s_t^\star(a) = \tau_a(X_t) - \lambda_t c(X_t,a)$, and let $a_t^\star$ maximize this score over $\mathcal{A}_t^{\mathrm{aff}}$. On the confidence event of Lemma G.3, no action can eliminate $a_t^\star$, so $a_t^\star \in \mathcal{B}_t^{\mathrm{NTK}}$. The sampled action is one endpoint of the farthest pair and also lies in the active set. Its active-set inequality gives

$$\widehat{s}_t(a_t^\star) - \widehat{s}_t(A_t) \leq 2r_t(a_t^\star, A_t).$$

Because costs are known exactly, the pairwise confidence lemma directly gives

$$|\{s_t^\star(a_t^\star) - s_t^\star(A_t)\} - \{\widehat{s}_t(a_t^\star) - \widehat{s}_t(A_t)\}| \leq r_t(a_t^\star, A_t).$$

These are a single pairwise error and the active-set error, rather than two separate endpoint errors. Hence

$$s_t^\star(a_t^\star) - s_t^\star(A_t) \leq 3r_t(a_t^\star, A_t).$$

By the farthest-pair definition,

$$\|g_0(a_t^\star) - g_0(A_t)\|_{(V_{t-1}^0)^{-1}} \leq \|d_t^0\|_{(V_{t-1}^0)^{-1}}.$$

For a singleton active set, both sides are zero; otherwise the inequality follows from maximality over distinct pairs. Since $\|d_t^0\|_2 \leq 2L_g$ and $V_{t-1}^0 \succeq \alpha_{\mathrm{NTK}}I$, $\|d_t^0\|^2_{(V_{t-1}^0)^{-1}} \leq 1$. The elliptical-potential identity gives

$$\sum_{t=1}^T \|d_t^0\|^2_{(V_{t-1}^0)^{-1}} \leq 2\log\frac{\det V_T^0}{\det(\alpha_{\mathrm{NTK}}I)} = 2\Gamma_T^0.$$

Cauchy-Schwarz and monotonicity of the radius therefore yield

$$\sum_{t=1}^{T}\{s_t^\star(a_t^\star) - s_t^\star(A_t)\} \le 3\beta_T^{\mathrm{NTK}}\sqrt{2T\Gamma_T^0} + 3\epsilon_{m,J}T. \tag{G.18}$$

This is the neural online term. Through Eq. (G.12), it includes the statistical information gain, the NTK RKHS norm, and the finite-width errors. Assumption G.2 gives $\Gamma_t^0 \le \overline{\Gamma}_T^0$ and hence $\beta_t^{\mathrm{NTK}} \le \overline{\beta}_T$ for every $t \le T$.

It remains to control the budget terms. If $\lambda_t > \overline{\lambda}_T$, then for every priced action a,

$$\widehat{s}_t(a_0) - \widehat{s}_t(a) = \lambda_t c(X_t, a) - h(a, X_t; \theta_{t-1}) > 2r_t(a_0, a).$$

Thus all priced actions are eliminated, the algorithm plays a_0, and the queue decreases. Any other round raises the queue by at most $\eta(c_{\max} + \rho)$. Induction proves $0 \le \lambda_t \le \Lambda_T$. The standard squared-queue calculation gives

$$-\sum_{t=1}^{T}\lambda_t\{c(X_t, A_t) - \rho\} \le \frac{\lambda_1^2}{2\eta} + \frac{\eta T(c_{\max} + \rho)^2}{2}.$$

The nonnegative queue update also gives the deterministic prefix bound

$$\sum_{s\le t} c(X_s, A_s) \le t\rho + \frac{\Lambda_T}{\eta}.$$

Before round t, this bound implies

$$B - S_{t-1} \ge (T - t + 1)\rho - \frac{\Lambda_T}{\eta}.$$

Whenever the right-hand side is at least $c_{\max}$, every action is affordable. By the definition of N_T^{aff}, restricted affordability can therefore occur only during the final N_T^{aff} rounds. The free control is always affordable and has adjusted score zero. Replacing the global Lagrangian maximizer by the affordable maximizer thus costs at most R_{NTK} on each restricted round and at most $R_{\mathrm{NTK}}N_T^{\mathrm{aff}}$ in total.

Finally, define

$$\Phi_t(a, \lambda) := \tau_a(X_t) - \lambda\{c(X_t, a) - \rho\},$$
$$\mathcal{L}(\lambda) := \mathbb{E}_X\left[\max_{a\in\mathcal{A}}\{\tau_a(X) - \lambda(c(X, a) - \rho)\}\right].$$

Weak duality gives $\mathrm{OPT}_{\mathrm{LP}}(\rho) \le \mathcal{L}(\lambda_t)$. Since X_t is independent of $\mathcal{H}_{t-1}^{\mathrm{N}}$ and λ_t is $\mathcal{H}_{t-1}^{\mathrm{N}}$-measurable,

$$Z_t := \mathcal{L}(\lambda_t) - \max_{a\in\mathcal{A}}\Phi_t(a, \lambda_t)$$

is a martingale difference. Define the stopping time $\zeta = \inf\{t : \lambda_t > \Lambda_T\}$, with $\inf\varnothing = T + 1$, and the stopped martingale

$$\mathcal{M}_T^\zeta := \sum_{t=1}^{T}\mathbf{1}\{t < \zeta\}\left\{\mathcal{L}(\lambda_t) - \max_{a\in\mathcal{A}}\Phi_t(a, \lambda_t)\right\}$$

whose increments are bounded in absolute value by the deterministic quantity $2\{R_{\mathrm{NTK}} + \Lambda_T(c_{\max} + \rho)\}$. Because λ_t is $\mathcal{H}_{t-1}^{\mathrm{N}}$-measurable and Λ_T is deterministic, the stopped process is adapted to the original filtration; its concentration is not conditioned on $\mathcal{E}_{\mathrm{cpl}}$. Let $\mathcal{Z}_\delta$ be its Azuma-Hoeffding event. Then $\mathbb{P}(\mathcal{Z}_\delta) \ge 1 - \delta$. On $\mathcal{E}_{\mathrm{cpl}}$, the queue bound above implies $\zeta = T + 1$, so $\mathcal{M}_T^\zeta = \sum_{t=1}^{T} Z_t$. The identity $\tau_{A_t}(X_t) = \Phi_t(A_t, \lambda_t) + \lambda_t\{c(X_t, A_t) - \rho\}$ therefore gives

$$\begin{aligned} R(T) \le &\sum_{t=1}^{T}\left\{\max_{a\in\mathcal{A}}\Phi_t(a, \lambda_t) - \Phi_t(A_t, \lambda_t)\right\} \\ &- \sum_{t=1}^{T}\lambda_t\{c(X_t, A_t) - \rho\} + \mathcal{M}_T^\zeta. \end{aligned} \tag{G.19}$$

On $\mathcal{Z}_\delta$, the last term is bounded by the fourth term of Eq. (G.17). Combining this event with Lemma G.3, Eq. (G.18), the deterministic dual bound, and the affordable-set comparison proves the result. The definition of the affordable set gives pathwise feasibility. □

Corollary G.5 (Effective-dimension form)**.** *Under the hypotheses of Theorem G.4, suppose L_g, R_h, R_{NTK}, α_{NTK}, S_{NTK}, Ξ, η_0, ρ, $c_{\min}^{-1}$, and $c_{\max}$ are controlled independently of T. On $\mathcal{E}_{\mathrm{cpl}} \cap \mathcal{C}_\delta \cap \mathcal{Z}_\delta$, which has probability at least $1 - 2\delta - \delta_{\mathrm{cpl}}$, Eq.* (G.17) *implies*

$$R(T) = \widetilde{O}\Big(\widetilde{d}_{\mathrm{ac}}\sqrt{T} + \epsilon_{\mathrm{init}}(m)T\sqrt{\widetilde{d}_{\mathrm{ac}}\log T} + \{\epsilon_{\mathrm{fw}}(m,T) + \epsilon_{\mathrm{opt}}(m,T,J)\}T\Big). \tag{G.20}$$

A sufficient set of finite-width and optimization rates for preserving the statistical order is

$$\epsilon_{\mathrm{init}}(m) = \widetilde{O}\left(\sqrt{\frac{\widetilde{d}_{\mathrm{ac}}}{T}}\right), \qquad \epsilon_{\mathrm{fw}}(m,T) + \epsilon_{\mathrm{opt}}(m,T,J) = \widetilde{O}\left(\frac{\widetilde{d}_{\mathrm{ac}}}{\sqrt{T}}\right).$$

Under these sufficient rates, the regret preserves the $\widetilde{O}(\widetilde{d}_{\mathrm{ac}}\sqrt{T})$ statistical order. The weaker conditions $\epsilon_{\mathrm{init}}(m) = o((\widetilde{d}_{\mathrm{ac}}\log T)^{-1/2})$ and $\epsilon_{\mathrm{fw}}(m,T) + \epsilon_{\mathrm{opt}}(m,T,J) = o(1)$, together with $\widetilde{d}_{\mathrm{ac}}\log T = o(\sqrt{T})$, suffice for sublinear regret at a fixed confidence level. The dimension condition ensures that the statistical contribution is itself $o(T)$.

Proof. Assumption G.2 and Eq. (G.15) imply $\Gamma_T^0 = O(\widetilde{d}_{\mathrm{ac}}\log T)$. Substitute this relation into Eqs. (G.12), (G.16), and (G.17). The statistical product $\sqrt{\Gamma_T^0}\sqrt{T\Gamma_T^0}$ is $O(\Gamma_T^0\sqrt{T})$. Initialization approximation contributes $O(\epsilon_{\mathrm{init}}T\sqrt{\Gamma_T^0})$, and direct nonlinear coupling contributes $O(\epsilon_{\mathrm{cpl}}T)$. Because $\widetilde{d}_{\mathrm{ac}} \geq 1$, the queue range is at most $\widetilde{O}(\sqrt{\widetilde{d}_{\mathrm{ac}}} + \epsilon_{\mathrm{init}}\sqrt{T} + \epsilon_{\mathrm{cpl}})$. Multiplying this quantity by $\sqrt{T}$ in the master-martingale and affordable-set terms is dominated by the three displayed orders in Eq. (G.20). This proves that equation; the two rate statements follow by direct comparison with $\widetilde{d}_{\mathrm{ac}}\sqrt{T}$ and T. □